Inferring the Aggregate: Information Frictions and Fiscal Policy

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Abstract

Households use their own experiences to form expectations about the macroeconomy. Changes in household income can thus affect decisions through both resources and information. I study the importance of this channel in a heterogeneous-agent model in which households infer aggregate movements from their own cash flows. When macroeconomic conditions are not fully observed, countercyclical transfers are not only resource windfalls. In addition, they signal to households that aggregate conditions are weak. Evidence from a randomized information experiment supports this mechanism: fiscal information lowers GDP forecasts and reduces durable spending intentions, with the decline in the latter strongest among high-income households. To solve the model, I develop a global method that combines Bayesian filtering with approximate aggregation. On impact, the information channel reduces the consumption response to a transfer from about 38 cents per dollar under full information to about 7 cents under dispersed information. As in the data, negative spending responses to information are strongest at the top of the distribution. Moreover, when households infer aggregate conditions from cash flows, they learn about aggregate shocks gradually, generating hump-shaped consumption responses. I find that targeting transfers toward low-income households can mitigate the effects of the information channel, both raising the impact consumption response and restoring a front-loaded aggregate response.

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1 Introduction

Households have imperfect information about the macroeconomy. Changes in household income can reflect both idiosyncratic circumstances and macroeconomic conditions, so realized cash flows contain information about aggregate variables. What households infer from these cash flows matters for their expected future income and spending. Observed cash flows therefore affect consumption through both resources and information. This paper studies how important this information channel is for household spending, particularly in response to countercyclical fiscal transfers.

I study this question by extending models of dispersed information, in which households form expectations rationally under imperfect information, to a heterogeneous-agent economy with endogenous aggregate output. Households face borrowing constraints and differ in income as well as wealth. They observe their own cash flows but not aggregate conditions directly. Given this information structure, rational expectation formation implies that households use these cash flows to infer aggregate conditions. The information structure has two main implications: First, households learn about persistent aggregate shocks only gradually, dampening the initial consumption response and generating hump-shaped dynamics. Second, cash flows affect consumption through revisions in future income expectations, an effect I refer to as the information channel. I find that this information channel can be negative in response to countercyclical transfers, partly offsetting their direct resource effect.

To motivate the information structure, I turn to a large empirical literature on expectation formation (Kuchler and Zafar, 2019; D'Acunto et al., 2021; Taubinsky et al., 2026), as well as my own analysis using the New York Fed Survey of Consumer Expectations (SCE). The literature broadly documents imperfect information and significant extrapolation by households from their own economic conditions to aggregate ones. In the SCE data, I show that households' financial conditions predict their expectations about aggregate conditions and provide evidence consistent with interpreting this relationship as rational inference from their own economic circumstances.

Countercyclical transfers provide a natural test of the importance of the information channel: They are salient, directly observed cash flows that respond to macroeconomic conditions. A countercyclical stimulus payment is both a positive shock to current resources and a negative signal about aggregate conditions, reducing consumption through the information channel. In general, households incorporate fiscal policy into their expectation formation. However, different policies appear to carry different information. News about procyclical or acyclical fiscal expansions leads to more optimistic short-run expectations

(Andre et al., 2022; Ludolph et al., 2026). In contrast, information about countercyclical policy can instead induce pessimism, with households interpreting the policy response as a signal that aggregate conditions are weak (Coibion et al., 2022; Melosi et al., 2025). Such pessimistic revisions could then depress household spending, offsetting part of the stimulus provided by the policy itself.

I provide empirical evidence for the importance of the information channel using the Bundesbank Online Panel – Households (BOP-HH), in which survey respondents were informed of a EUR 750 billion fiscal package in April 2020. Consistent with Metiu and Stockerl (2024), I show that information about the package lowers expected GDP over the next twelve months by 1.76 percentage points and the stated probability of large durable purchases over the same period by 6.15 percentage points relative to respondents that did not receive the information treatment. I extend this analysis by examining heterogeneity across income groups. GDP expectations fall broadly across income groups, while the decline in spending intentions is concentrated among high-income households. This is consistent with less constrained households adjusting consumption more strongly in response to the same information. These findings speak directly to the information channel central to the model mechanism.

I then build a partial-equilibrium model to illustrate the mechanism. In this model, heterogeneous households face uninsurable income risk and a borrowing constraint. Households do not directly observe aggregate productivity but observe their own wages and rationally infer the former from the latter. The first implication of the information structure is that households learn slowly about an increase in productivity. The consumption response to an aggregate shock becomes hump-shaped. The second implication is that a wage increase now raises a household's expectations about aggregate productivity, with the household rationally inferring that aggregate conditions may be strong.

To quantify the importance of information frictions for the efficacy of fiscal transfers, I extend the model to a quantitative general-equilibrium heterogeneous-agent setting with a demand externality. This framework retains the household structure from the partial-equilibrium model. Fiscal policy operates through direct transfers to households and contains a discretionary as well as a countercyclical component. Government debt absorbs the initial financing needs, while lump-sum taxes respond to stabilize debt over time. The model is non-Ricardian, with borrowing constraints and incomplete markets implying that the timing of transfers and taxes can affect consumption. The demand externality introduces general-equilibrium feedback from household spending to output and wage income similar to that in heterogeneous-agent New Keynesian models. Households observe their own wages, but cannot perfectly separate aggregate from idiosyncratic

movements. Additionally, households observe the fiscal transfers received, but cannot distinguish perfectly between discretionary and countercyclical components. The two implications from the basic model carry over, although information is now also conveyed through transfers: If fiscal policy responds to falling output with transfers, the transfers themselves become a signal, with households inferring that aggregate conditions may be weak when receiving them.

The full quantitative model features heterogeneous agents facing borrowing constraints, dispersed information, and demand-determined aggregate output. Solving such models requires jointly determining individual policies, expectations, and the infinite-dimensional cross-sectional distribution over household assets, incomes, and beliefs. This poses a substantial computational challenge under dispersed information: Bayesian filtering can only proceed when the underlying state process is known. But in this class of models, that very process depends on households' decisions, which in turn depend on their expectations. Demand-determined output makes the computational challenge particularly severe because signals are contemporaneous, endogenous variables. Incomes are signals and affect households' expectations, expectations affect their decisions, and these decisions then determine incomes through general equilibrium feedback.

Because existing methods do not directly apply to this setting, I develop a solution method combining Bayesian filtering and approximate aggregation to solve approximately for an equilibrium. The approximation enters through a parametric perceived law of motion. Conditional on this law of motion, households form expectations optimally using Bayesian filtering. Starting from an initial guess, the coefficients are iteratively updated until the perceived law of motion accurately matches the simulated aggregate dynamics of the model. The simulation involves solving for market clearing aggregate variables consistent with the signals observed by households. I find the approximation of aggregate dynamics to be highly accurate. At the household level, the Bayesian filter also tracks aggregate output more accurately than alternative machine-learning methods while requiring far fewer household state variables. I also extend the method to accommodate non-Gaussian signals arising from transfer eligibility, using an approximation to household posteriors.

Information frictions have a sizable impact on the response of consumption to fiscal stimulus. Quantitatively, the frictions reduce the initial aggregate consumption response from roughly 38 cents per dollar under full information to approximately 7 cents. The departure from full information thus substantially reduces the efficacy of stimulus policies. The direct resource effect is similar across information structures, while the negative news conveyed by the transfer accounts for the decrease. The importance of the information channel increases with cash on hand. For households at the top of the distribution, the

information channel more than offsets the positive resource effect, and consumption falls upon receiving a transfer. This pattern follows from differences in responsiveness to information, with less constrained households adjusting present spending more strongly in response to deteriorating expectations about future income. This matches the experimental evidence from the BOP-HH data, providing cross-sectional support for the key mechanism. Beyond these effects on the initial response, gradual learning dampens consumption responses to persistent aggregate shocks and generates hump-shaped dynamics.

These results also help account for modest household and aggregate consumption responses to countercyclical stimulus payments, partially reconciling high micro MPCs from windfall cash receipts with modest macro responses (Orchard et al., 2025; 2026).[1] High MPCs from other cash receipts need not carry over to stimulus payments when the latter also convey information about aggregate conditions.

I also study how targeting transfers toward low-income households changes these results. Targeting in this way redirects resources to households with higher spending responses already in a full information framework. Under dispersed information, targeting additionally reallocates transfers toward households that are less sensitive to the informational channel. Due to this additional channel, I find that the gains from targeted transfers relative to untargeted ones are larger under dispersed information. With untargeted transfers, the impact consumption response under full information is more than five times that under dispersed information. With targeting, it is only about 30 percent larger, closing most of the gap. Through the reallocation to less information-sensitive households, targeting also restores a front-loaded aggregate consumption response.

The paper makes three contributions. First, it identifies and empirically disciplines an information channel of fiscal transfers. Second, it incorporates dispersed information and rational inference from endogenous cash-flow signals into a heterogeneous-agent model, capturing the heterogeneity in consumption responses to information observed in the data. Third, it develops a global method for computing approximate equilibria in this class of models.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents empirical evidence on household expectation formation and the information effects of fiscal policy. Section 4 illustrates the information mechanism. Section 5 introduces the quantitative fiscal-policy model and its solution method. Section 6 evaluates the model against the information experiment and presents the main quantitative results. Section 7 extends the analysis to targeted transfers. Section 8 concludes.

[1] See Appendix A for a detailed comparison.

2 Related literature

Expectations about aggregate variables are widely documented to be heterogeneous, and a large empirical literature links this heterogeneity to households' own experiences. Kuchler and Zafar (2019) show that households extrapolate from locally experienced house-price changes when forming expectations about future house-price growth. Cavallo et al. (2017) and D'Acunto et al. (2021) document that inflation expectations respond to the prices households themselves observe, especially grocery prices. Weber et al. (2022) survey this evidence. Link et al. (2026) further link personal experiences to attention to the macroeconomy and inflation expectations. Taubinsky et al. (2026) show how these expectations respond to other idiosyncratic shocks such as income. Closest to the mechanism here, Adams and Rojas (2024) find that households cannot separate idiosyncratic from aggregate components of their own income, such that purely idiosyncratic income shocks move expectations about aggregates. Motivated by this evidence, this paper models households as observing variables entering their own budget constraints and treating them as signals about aggregate equilibrium objects, which they process through Bayesian updating.

There is a large literature bringing dispersed information into theoretical models. One strand of this literature focuses on explicitly linear models, or ones that are linearized as part of the solution. The seminal work is by Lucas (1972), who studies monetary non-neutrality in an island economy where agents cannot perfectly distinguish aggregate from local shocks. Relatedly, Lorenzoni (2009) studies dispersed signals and demand shocks in a business-cycle model, while Angeletos and La'O (2020), Kohlhas and Walther (2021), and related work analyze how dispersed information influences expectations and aggregate dynamics. Nimark (2017) and Huo and Takayama (2025) develop methods for solving linear rational expectations models with higher-order beliefs.

Another strand of the theoretical literature relaxes the rational expectations assumption. In addition to generating a variety of results, this allows researchers to tackle more complex or nonlinear models. Closely related to the dispersed information literature discussed previously is work by Bordalo et al. (2018), Angeletos et al. (2021), and Guarda (2026). There is also a large literature that deviates from rational expectations by considering subjective beliefs. Marcet and Sargent (1989) and Evans and Honkapohja (2001) implement this through adaptive learning about the underlying model process. A separate strand considers "internal rationality" based on work by Adam and Marcet (2011). Adam et al. (2016) as well as Adam et al. (2017) use this concept to implement heterogeneity in expectations. Relatedly, Kurz et al. (2013) study aggregation in a New Keynesian model with heterogeneous beliefs under common information. In these approaches, heterogeneity

in expectations can arise from subjective beliefs about the underlying process, even when agents share the same information.

The paper also relates to work on sluggish expectations and excess smoothness. Sticky-information models, including Mankiw and Reis (2002), Carroll (2003), Carroll et al. (2020), and Auclert et al. (2020), generate slow-moving expectations by modeling agents that update their information sets only occasionally. Maćkowiak and Wiederholt (2015) also generate sluggish consumption responses through rational inattention. Campbell and Deaton (1989) study excess smoothness in aggregate consumption, while Dynan (2000) tests habit formation and finds no evidence of habits at the annual frequency. In this paper, expectations adjust slowly because households learn gradually from dispersed signals about aggregates, rather than because updating is infrequent, attention is costly, or preferences feature habit formation. An early contribution connecting incomplete aggregate information and inference from individual income to excess smoothness is Pischke (1995).

The fiscal information channel in this paper relates to evidence that expansionary policy announcements can induce pessimistic revisions in expectations. Metiu and Stockerl (2024), on whose work this paper builds, use the same BOP-HH experiment and document the increase in pessimism following information about the fiscal package. Evidence from fiscal announcements in Japan also supports an adverse information channel, particularly when unexpectedly large stimulus announcements signal weak economic conditions (Melosi et al., 2025). Related evidence comes from Coibion et al. (2022), who find that informing households about the fiscal response to the COVID-19 pandemic raises short-run unemployment expectations and lowers expected household income growth. More generally, Roth and Wohlfart (2020), D'Acunto et al. (2022), and Coibion et al. (2023) provide evidence that households' macroeconomic expectations affect their spending decisions. This paper connects such information effects to household consumption decisions and quantifies their implications for countercyclical transfers.

At the household level, there is significant evidence that cash receipts induce strong consumption responses (Souleles, 1999; Kueng, 2018). Selected estimates of short-run MPCs out of stimulus payments are smaller than these non-stimulus benchmarks (Orchard et al., 2025, 2026; Parker et al., 2022). During COVID-19, Goldfayn-Frank et al. (2022) also find a modest household spending response to the German *Kinderbonus*, with an overall MPC of 5.4% across its three payments. These comparisons are suggestive, since consumption measures, payment design, recipient characteristics, and estimation methods differ across studies. Appendix A.1 presents a suggestive comparison of selected estimates and discusses differences in payment size, spending horizons, and policy context. The

information mechanism in this paper provides a reason why the countercyclical policy context itself can reduce household spending responses.

At the aggregate level, Taylor (2009, 2022) questions the effectiveness of stimulus payments in raising consumption. Orchard et al. (2025) find that the 2008 rebate payments had modest effects on aggregate consumption, driven by lower micro MPC estimates as well as general-equilibrium crowding out. Ramey (2026) documents limited aggregate effects of temporary cash transfers across several developed economies. Appendix A.2 illustrates these limited aggregate effects. My analysis complements this evidence by developing and quantifying an information channel through which countercyclical transfers can induce pessimistic revisions in expectations and lower aggregate consumption responses.

Quantitative heterogeneous-agent models also emphasize the importance of transfer design. Bayer et al. (2023) show that targeting transfers based on unemployment can induce large responses, while untargeted transfers may generate responses near zero. McKay and Reis (2016) also highlight the importance of insurance through transfers, while Oh and Reis (2012) find that redistribution matters for their effects. This paper studies an additional channel arising when aggregate conditions are imperfectly observed: transfers may convey information about the state of the economy, and the resulting revisions in expected future income can weaken their effects. Among my findings is that targeting becomes even more important for fiscal effectiveness when this information channel is taken into account.

Additionally, the present paper relates to work on information transmission through fiscal policy, for example by Melosi et al. (2025) as well as Georgarakos and Kenny (2022), and to the large literature on the signaling effects of monetary policy, including Melosi (2017), Nakamura and Steinsson (2018), Kohlhas (2022) as well as Müller et al. (2024).

The main method used to solve the model is an extension of the one developed by Krusell and Smith (1998). In line with work by Marcet and Sargent (1989), this can be viewed as solving for a restricted perceptions equilibrium (RPE) which approximates the full rational expectations equilibrium (REE). Similar approaches are used by Porapakkarm and Young (2008) as well as Guarda (2026) to solve Krusell-Smith economies in which households face dispersed information about an endogenous capital stock. In a similar framework, Broer et al. (2026) employ approximate aggregation in a model with endogenous information acquisition. The method developed here extends these approaches to a setting with contemporaneous feedback between household decisions and information: current consumption affects aggregate output and hence the signals from which households form expectations. My method also relates closely to Hellwig and Venkateswaran (2011, 2025), who address a similar computational problem in a model of firm price-setting.

On the general topic of solving heterogeneous-agent models with dispersed information, this paper is closely related to ongoing work by Auclert et al. (2026) and Overhage (2026), who develop differing approaches to solving this class of models in sequence space. Both of these papers solve for local, linear dynamics around a full-information steady state. Relative to these, the solution method in this paper is global and approximates aggregate dynamics through a perceived law of motion, conditional on which households update optimally using Bayesian filtering.

Lastly, this paper relates to Moll (2026), who proposes replacing rational expectations with beliefs over a more limited set of aggregate variables. One of his proposed solutions is to model expectation formation in line with survey evidence. The present paper demonstrates that an information friction can microfound this even in an approximate rational expectations equilibrium, and proposes a tractable solution method for it.

3 Empirical evidence

3.1 Experience effects and forecast efficiency

The SCE asks respondents whether their household is financially better or worse off than twelve months earlier. The response, denoted by $Q1_{it}$, is recorded on a five-point scale from 1, much worse, to 5, much better. Respondents also report the percent chance that the aggregate unemployment rate will be higher twelve months later, denoted by $Q4new_{it}$.

Figure 1 plots weighted mean unemployment expectations across the five financial-situation categories. Respondents who report that their household is financially worse off assign a substantially higher probability to an increase in aggregate unemployment. The relationship is monotonic across the financial-situation categories.

To quantify this relationship, I estimate

$$Q4new_{it} = \alpha_i + \delta_t + \beta Q1_{it} + \varepsilon_{it}, \tag{1}$$

where α_i denotes respondent fixed effects and δ_t denotes survey-month fixed effects. Month fixed effects absorb aggregate conditions and public information shared by respondents interviewed in the same month. Respondent fixed effects absorb persistent differences in beliefs and reporting behavior. The main estimates use SCE survey weights, with an unweighted specification reported as a robustness check. Standard errors are clustered by respondent.

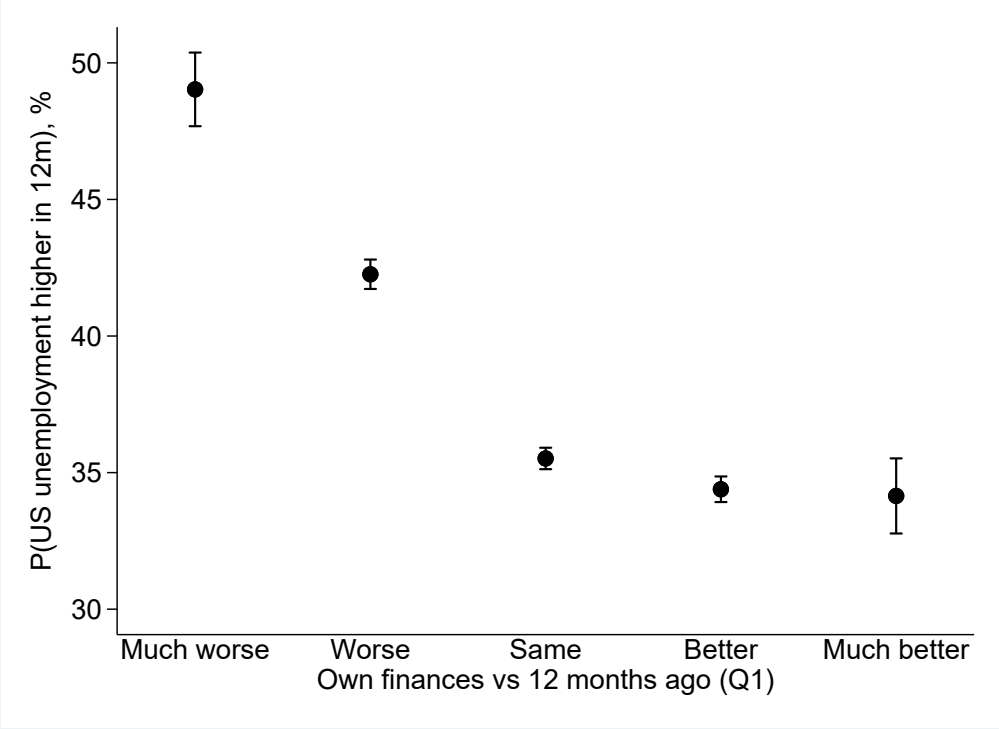


Figure 1: Aggregate unemployment expectations by household financial situation

The full fixed-effects ladder, including the pooled and respondent-fixed-effects-only specifications, with weighted and unweighted estimates, is reported in Appendix B.2. The survey-weighted specification with month fixed effects isolates the within-month relationship. A one-category improvement in the household's financial situation is associated with an approximately 3.8 percentage point decline in the reported probability that unemployment will rise. Because this comparison is made within survey month, the estimate cannot be explained by financially worse-off households simply being observed during periods of weak aggregate conditions.

When respondent fixed effects are added to this specification, the coefficient falls to approximately -1.6 percentage points but remains statistically significant. The smaller within-respondent estimate is consistent with the short length of the SCE panel and the limited variation in respondents' broad financial-status categories during their time in the survey.

These results imply that households interviewed at the same time do not hold identical expectations about a purely aggregate outcome. Personal financial circumstances predict disagreement even after conditioning on common aggregate information. This is consistent with dispersed information and experience-based inference.

I obtain complementary evidence from direct measures of household income. In the BOP-

Table 1: Own financial situation and expected aggregate unemployment

	P(unemployment higher)
No fixed effects	-3.94***
	(0.16)
Month fixed effects	-3.76***
	(0.16)
Respondent + month fixed effects	-1.60***
	(0.13)
Unweighted, respondent + month FE	-1.58***
Observations	185,086

HH, declines in household income are associated with lower expectations of aggregate growth and higher expected aggregate unemployment, but only the latter relationship is statistically significant. The point estimates suggest larger responses to real-income losses than to gains, although formal evidence for asymmetry is limited. This is likely due to much of the data being collected during times of high inflation, in which nominal income increases do not necessarily imply real ones. Appendix B.2 reports the full BOP-HH income specification.

The SCE relationship between personal finances and aggregate beliefs does reject full information, but not necessarily rational expectations. Households may hold different forecasts because they observe different signals, while still using those signals efficiently. I therefore compare the relationship of financial conditions with respondents' forecasts and with subsequent aggregate realizations. Following Taubinsky et al. (2026), this can be interpreted as a test of rational expectations updating under imperfect information.

Define the forecast as

$$F_{it} = Q4new_{it}/100, \tag{2}$$

and the realization as

$$R_{it} = \mathbf{1}\{u_{t+12} > u_t\}, \tag{3}$$

where u_t is the US unemployment rate in the interview month. The forecast error is

$$FE_{it} = R_{it} - F_{it}. \tag{4}$$

For each outcome $d_{it} \in \{F_{it}, R_{it}, FE_{it}\}$, I estimate

$$d_{it} = \alpha + \beta_d Q1_{it} + \varepsilon_{it}. \tag{5}$$

The regressions use SCE survey weights and two-way clustering by survey month and respondent. Clustering by month is important because the realized outcome is common to all respondents interviewed in the same month. Respondent clustering allows forecast and reporting errors to remain correlated within individuals.

Table 2: Forecast efficiency with respect to household finances

	Slope on own finances
Forecast	-0.0391***
	(0.0021)
Realization	-0.0362***
	(0.0105)
Forecast error (realized minus forecast)	0.0029
	(0.0108)
Observations	180,863

A one-category improvement in financial conditions is associated with a 3.9 percentage point decline in the forecasted probability of higher unemployment. The corresponding realization slope is a 3.6 percentage point decline. The forecast-error slope is 0.3 percentage points and is not statistically distinguishable from zero. Appendix B.2 reports the corresponding estimates under alternative one-way clustering conventions.

The similarity of the forecast and realization coefficients implies that the differences in unemployment expectations across financial-situation categories broadly reflect differences in the probability of unemployment subsequently increasing. This is an efficiency result rather than evidence of full information. Households may hold different forecasts because they possess different information, while their forecast errors remain orthogonal to that information. Taken together, the evidence is consistent with rational updating under dispersed information.

3.2 The fiscal-package signal

I next examine whether a fiscal-policy announcement conveys information about aggregate conditions and affects household spending plans. The analysis uses a randomized information experiment conducted in the April 2020 wave of the BOP-HH (Deutsche Bundesbank, 2020). A detailed discussion of this experiment is provided by Metiu and Stockerl (2024). Respondents were assigned to one of six information treatments. The analysis in this section compares respondents receiving information about the German federal fiscal package, arm 5, with respondents receiving a placebo text about European human rights, arm 1. The remaining experimental arms are excluded. Covariate balance for the fiscal-package and placebo groups is reported in Appendix B.3.

Immediately after treatment, respondents report a probability distribution over German GDP movements during the following twelve months. I convert this distribution into a mean growth forecast using the midpoints of the eight response bins. Respondents also report propensities from 0 to 100 to purchase a car, purchase another large durable good, and build up additional financial reserves over the following twelve months.[2]

For the effect on GDP expectations, I estimate

$$GDP_i^e = \alpha + \beta fp_i + X_i'\lambda + \varepsilon_i, \tag{6}$$

where fp_i equals one for the fiscal-package treatment and zero for the placebo treatment. The vector X_i contains gender, age, household size, employment status, and pre-treatment growth expectations. The regression uses BOP-HH survey weights and heteroskedasticity-robust standard errors.

[2]Savings-related outcomes and their interpretation are discussed in Appendix B.7.

Table 3: Policy information and expected GDP growth

	Expected GDP growth
Fiscal package (arm 5)	-1.76***
	(0.56)
Adjusted placebo prediction	-1.86
Unweighted estimate	-1.37***
Controls	Yes
Observations	628

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

The fiscal-package announcement lowers expected GDP relative to the placebo treatment. Respondents therefore interpret the announcement not only as information about the government's policy response, but also as information that underlying aggregate conditions are weak. It should be noted that identifying this (adverse) information channel is not this paper's contribution, as this effect was already identified by Metiu and Stockerl (2024). Overall, informing households about the size of the fiscal package induces a similar degree of pessimism as informing them about general economic conditions. Figure 23 in Appendix B.3 places this effect in the context of all six experimental arms. Additionally, Appendix B.5 shows that the effect on expectations is persistent, although there is significant attrition in the panel over subsequent survey waves.

The increase in pessimism about future GDP appears in line with other expectations indicators, with exploratory results suggesting that treated households are more likely to identify the economy as a serious problem or report concerns about their finances going forward. Results regarding these additional indicators are collected in Appendix B.1.3.

I next examine whether the treatment affects intended durable spending differently across the household-income distribution. Figure 2 reports treatment effects separately for households with monthly income below EUR 5,000 and households at or above EUR 5,000.

The fiscal-package treatment has little effect on intended durable spending among households below the threshold but produces a substantially larger decline among households with income of at least EUR 5,000. The difference suggests that the same negative aggregate signal produces heterogeneous spending responses, even among groups whose beliefs move in broadly similar directions.

A natural interpretation is that higher-income households have greater scope to adjust

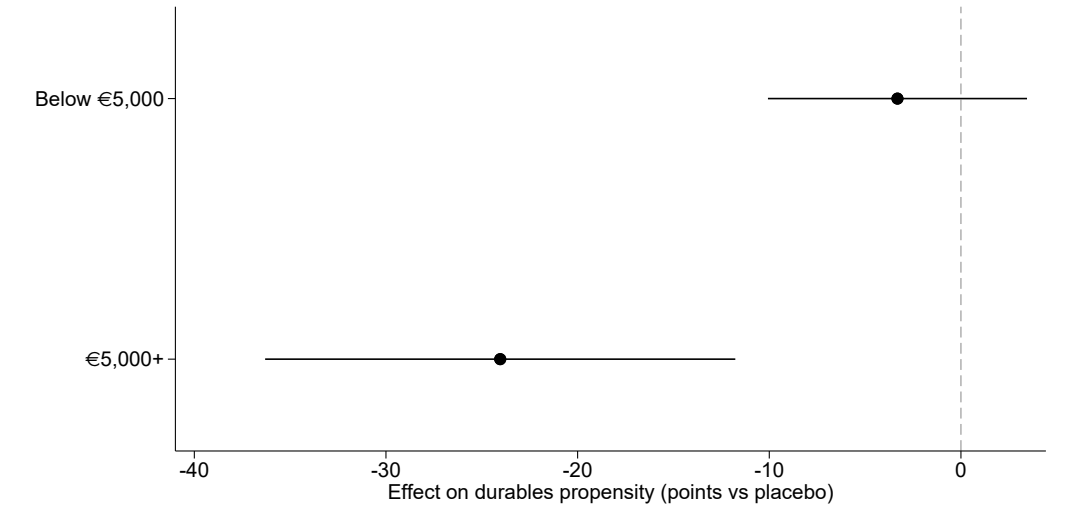


Figure 2: Fiscal-package information and intended durable spending by household income

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

discretionary durable purchases in response to negative news. The result is therefore consistent with a common information update combined with heterogeneous marginal spending responses. Appendix B.4 reports the income-bracket profiles.

The belief effect also remains negative in subsequent BOP-HH waves, although the estimates become less precise as the experimental panel attrits. Figure 25 in Appendix B.5 reports the follow-up results.

Overall, the experiment supports the interpretation of fiscal policy as both an economic intervention and an information signal. Information about the fiscal package lowers expected growth and reduces intended spending among households with substantial capacity to adjust durable purchases. Existing empirical evidence likewise suggests that household resources matter for spending responses to changes in expectations. Lieb and Schuffels (2022) find related evidence for inflation expectations, while Yin (2026) documents the importance of financial constraints for responses to credit-related information. These findings motivate a heterogeneous-agent model in which financial resources affect the consumption response to information.

4 Information mechanism

This section uses a simple partial-equilibrium consumption–savings environment to isolate the information mechanism underlying the quantitative model below. Households infer aggregate productivity from their own income, which contains both an aggregate and an idiosyncratic component. The setup highlights two implications. First, cash flows affect consumption through both resources and information. Second, gradual learning about aggregate shocks can generate delayed aggregate responses.

4.1 Setup and signal extraction

Time is discrete, $t = 0, 1, 2, \ldots$. There is a unit mass of identical firms and a unit mass of households. Labor supply is inelastic and normalized to one. Firms produce with linear technology

$$Y_t = Z_t, \tag{7}$$

where, denoting log variables using lowercase letters, aggregate productivity follows

$$z_{t+1} = \rho_z z_t + \mu_{t+1}, \qquad \mu_{t+1} \sim \mathcal{N}(0, \sigma_\mu^2). \tag{8}$$

Under perfect competition and constant returns, a common wage $W_t = Z_t$ would reveal the aggregate state exactly. Instead, I introduce wage scrambling following the idea of Boppart et al. (2024). Individual wages are dispersed deviations from aggregate productivity:

$$\int_0^1 W_{it}\, di = Z_t, \qquad w_{it} = z_t + \xi_{it} - \frac{\sigma_\xi^2}{2}, \qquad \xi_{it} \sim \mathcal{N}(0, \sigma_\xi^2). \tag{9}$$

The adjustment by $-\sigma_\xi^2/2$ ensures that average wages remain equal to Z_t.

Households have CRRA preferences,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{C_{it}^{1-\sigma}}{1-\sigma}, \tag{10}$$

and access to a storage technology with exogenous gross return R^*. Their budget constraint is

$$C_{it} + A_{i,t+1} = R^* A_{it} + W_{it}, \qquad A_{i,t+1} \geq \bar{a}, \qquad \bar{a} = 0. \tag{11}$$

Households do not observe Z_t directly. They observe their own wage and understand the

processes for Z_t and ξ_{it}. Defining

$$s_{it} \equiv w_{it} + \frac{\sigma_\xi^2}{2}, \tag{12}$$

the state and signal processes are

$$z_{t+1} = \rho_z z_t + \mu_{t+1}, \qquad \mu_{t+1} \sim \mathcal{N}(0, \sigma_\mu^2), \tag{13}$$

$$s_{it} = z_t + \xi_{it}, \qquad \xi_{it} \sim \mathcal{N}(0, \sigma_\xi^2). \tag{14}$$

Individual income therefore provides a noisy signal about aggregate productivity.

The system is linear and Gaussian, so households apply the Kalman filter. Let $\hat{z}_{i,t|t-1}$ denote household i's prior estimate of z_t. After observing s_{it},

$$\hat{z}_{i,t|t} = (1 - k_t)\hat{z}_{i,t|t-1} + k_t s_{it}, \tag{15}$$

$$k_t = \frac{P_{t|t-1}}{P_{t|t-1} + \sigma_\xi^2}, \tag{16}$$

where $P_{t|t-1}$ is the prior variance. The posterior variance follows the standard Kalman recursion and converges to a common stationary value. Because $k_t \in (0, 1)$, households only partially incorporate an income realization into their estimate of the aggregate state. Beliefs therefore adjust gradually to aggregate shocks.

Define cash on hand as

$$M_{it} = R^* A_{it} + W_{it}. \tag{17}$$

Once the posterior variance has converged, the household problem can be written using $(M_{it}, \hat{z}_{i,t|t})$ as individual states:

$$V(M_{it}, \hat{z}_{i,t|t}) = \max_{A_{i,t+1} \geq \bar{a}} \left\{ u(C_{it}) + \beta \mathbb{E}_{it} \left[V(M_{i,t+1}, \hat{z}_{i,t+1|t+1}) \right] \right\}, \tag{18}$$

where

$$C_{it} = M_{it} - A_{i,t+1}, \tag{19}$$

$$M_{i,t+1} = R^* A_{i,t+1} + W_{i,t+1}. \tag{20}$$

An observed income change therefore affects consumption through current cash on hand as well as through beliefs about future income.

4.2 Consumption implications

The information friction has two closely related implications for consumption dynamics. Aggregate shocks are incorporated into household beliefs only gradually, generating delayed aggregate adjustment. At the household level, the response to an observed income change depends not only on the additional resources it provides, but also on what it reveals about the aggregate state.

Figure 3 illustrates the first mechanism.

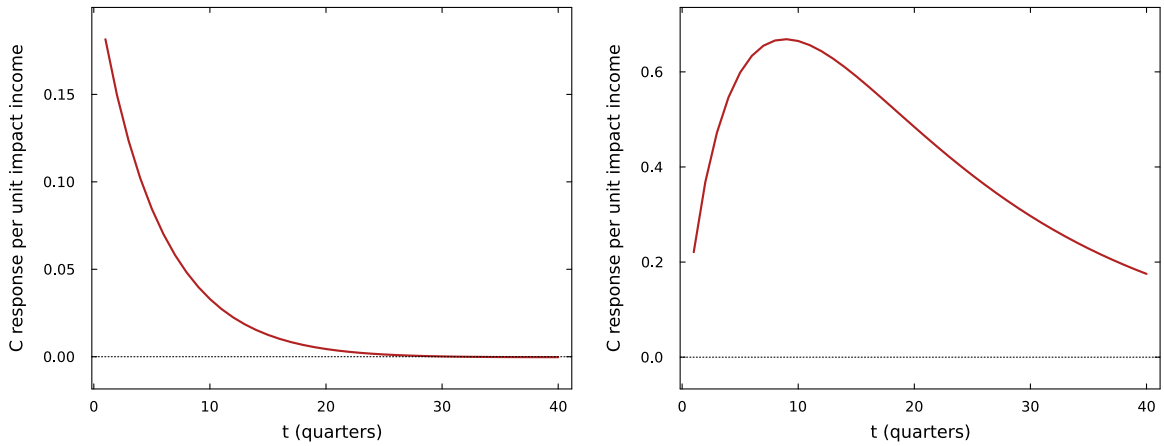


Figure 3: Consumption responses to one-standard-deviation shocks.

Notes: Consumption responses per unit of impact income; periods are quarters.

The left panel shows the consumption response of an individual household to a one-standard-deviation idiosyncratic wage shock, evaluated at median cash on hand. This response is front-loaded. The right panel shows the response of aggregate consumption to a one-standard-deviation aggregate productivity shock. On impact, consumption rises by about one-third of its eventual peak, with the maximum effect occurring only several quarters later.

The contrast reflects signal extraction. Aggregate shocks must first pass through dispersed income signals and Bayesian updating before they are fully incorporated into household beliefs, so their effect on aggregate consumption is spread over time. An idiosyncratic income innovation instead changes household resources immediately. This generates a front-loaded individual response alongside a delayed aggregate response.

The same mechanism implies that MPCs depend on the informational content of an income change. It is useful to distinguish between a pure cash-flow effect, which changes current resources while holding beliefs fixed, and an informational component operating through

the household's posterior belief about aggregate productivity.

Figure 4 illustrates this decomposition.

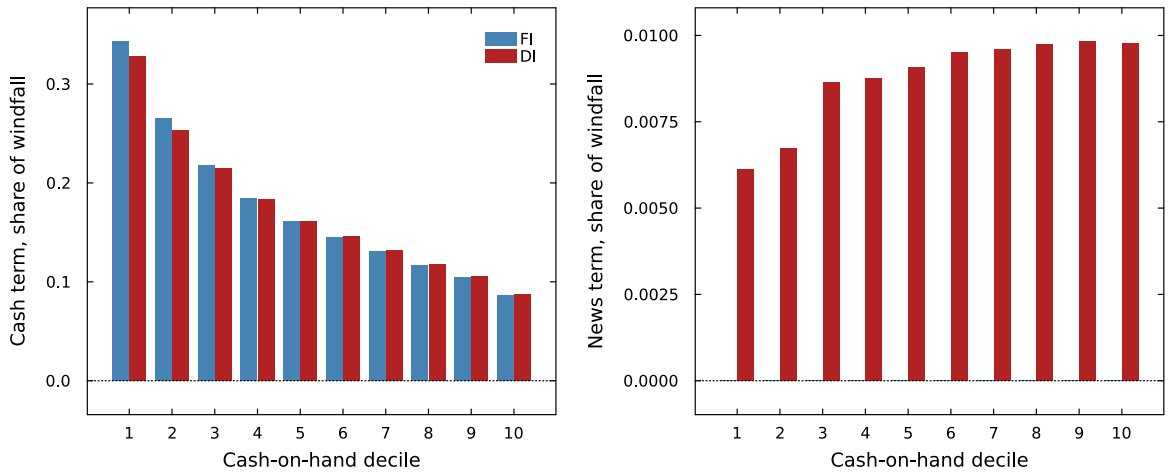


Figure 4: Quarterly responses to an idiosyncratic income shock

As in standard incomplete-markets models, the pure cash-flow effect is largest for households with little cash on hand and declines along the distribution. Under dispersed information, however, the same income realization also changes expectations. A positive wage innovation is partly interpreted as evidence that aggregate productivity, and therefore future income, is higher. The total consumption response can therefore differ from the pure MPC that holds beliefs fixed.

In this basic model, these effects are relatively minor overall. The information effect is less pronounced at the bottom of the distribution in absolute terms, and especially relative to the cash component. Constraints play the main role here, with households closer to the borrowing constraint less sensitive to expectations about future income. It is only at the top of the distribution that the information channel is about 10% as big as the cash component. While the size is small in this basic framework, the pattern of information mattering more at the top of the distribution holds true in the more involved model from Section 5 onward.

The environment therefore isolates the two mechanisms used in the fiscal-policy model below. Dispersed information delays aggregate adjustment because households learn about aggregate conditions over time, while the response to a cash flow depends on what that cash flow reveals about future economic conditions. In the fiscal-policy model, transfers themselves play this dual role as both resources and signals.

5 Fiscal-policy model

This section introduces fiscal transfers into the heterogeneous-household model with dispersed information and studies their quantitative effects. As in the basic model, households have fixed labor supply. Idiosyncratic income risk is now more general and allows for persistence. Transfers provide households with resources, but also convey information about aggregate conditions. Aggregate output responds to household consumption through a reduced-form demand externality, following, for example, Boppart et al. (2018) and Krueger et al. (2016).

Households observe the variables entering their own budget constraints, but cannot separate aggregate output from the idiosyncratic component of their labor income. They also observe aggregate transfers, but cannot determine whether a high transfer reflects a discretionary fiscal intervention or the systematic countercyclical response of fiscal policy. Households therefore use labor income and transfers to infer aggregate output, their own idiosyncratic efficiency, and the discretionary component of fiscal policy.

5.1 Environment

There is a unit mass of households indexed by $i \in [0, 1]$. Households save in an internationally traded asset with exogenous gross return

$$R^* = 1 + r^*, \qquad \beta R^* < 1. \tag{21}$$

Aggregate productivity follows the log AR(1) process

$$z_t = \rho_z z_{t-1} + \mu_t, \qquad \mu_t \sim \mathcal{N}(0, \sigma_\mu^2), \qquad |\rho_z| < 1, \tag{22}$$

where $z_t = \log Z_t$.

Households differ in persistent idiosyncratic efficiency. Let

$$e_{i,t+1} = \rho_e e_{it} + \zeta_{i,t+1}, \qquad \zeta_{i,t+1} \sim \mathcal{N}(\mu_\zeta, \sigma_\zeta^2), \qquad |\rho_e| < 1, \tag{23}$$

where $e_{it} = \log E_{it}$. The stationary variance of log efficiency is

$$\sigma_e^2 = \frac{\sigma_\zeta^2}{1 - \rho_e^2}. \tag{24}$$

I normalize mean efficiency in levels to one:

$$\mathbb{E}[E_{it}] = 1. \tag{25}$$

Since e_{it} is normally distributed, this requires

$$\mathbb{E}[e_{it}] = -\frac{\sigma_e^2}{2}. \tag{26}$$

The corresponding drift in the innovation process is

$$\mu_\zeta = -(1-\rho_e)\frac{\sigma_e^2}{2} = -\frac{\sigma_\zeta^2}{2(1+\rho_e)}. \tag{27}$$

With a continuum of households, the normalization implies

$$\int_0^1 E_{it}\, di = 1. \tag{28}$$

Household i has preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_{it}), \qquad u(C) = \frac{C^{1-\sigma}}{1-\sigma}. \tag{29}$$

Labor supply is fixed. Household i receives individual labor income W_{it}, an aggregate transfer T_t, and pays a lump-sum tax τ_t. Its budget constraint is

$$C_{it} + A_{i,t+1} = R^* A_{it} + W_{it} + T_t - \tau_t, \tag{30}$$

subject to

$$A_{i,t+1} \geq 0. \tag{31}$$

Cash on hand is therefore

$$M_{it} = R^* A_{it} + W_{it} + T_t - \tau_t. \tag{32}$$

Aggregate consumption is

$$C_t = \int_0^1 C_{it}\, di, \qquad c_t = \log C_t. \tag{33}$$

Aggregate output is determined by productivity and the aggregate consumption externality:

$$Y_t = Z_t C_t^\omega, \qquad \omega \geq 0. \tag{34}$$

Define $y_t = \log Y_t$. The production relation in logs is

$$y_t = z_t + \omega c_t. \tag{35}$$

Aggregate labor income equals output,

$$W_t = Y_t, \tag{36}$$

and individual labor income is

$$W_{it} = Y_t E_{it}. \tag{37}$$

Consequently, observed individual log labor income satisfies

$$w_{it} = y_t + e_{it}, \tag{38}$$

where $w_{it} = \log W_{it}$. When $\omega = 0$, aggregate output is determined entirely by productivity. When $\omega > 0$, higher household consumption raises output and labor income, which feeds back into household consumption.

5.2 Fiscal policy

Fix an output reference level Y^{ref} and define

$$y^{\text{ref}} = \log Y^{\text{ref}}, \qquad \widetilde{y}_t = \log\left(\frac{Y_t}{Y^{\text{ref}}}\right) = y_t - y^{\text{ref}}. \tag{39}$$

Normalized aggregate transfers are

$$\widetilde{T}_t = \frac{T_t}{Y^{\text{ref}}}. \tag{40}$$

Transfers respond countercyclically to output and contain a persistent discretionary component:

$$\widetilde{T}_t = -\psi_y \widetilde{y}_t + \nu_t, \qquad \nu_t = \rho_\nu \nu_{t-1} + \varepsilon_t^\nu, \qquad \varepsilon_t^\nu \sim \mathcal{N}(0, \sigma_\nu^2). \tag{41}$$

Equivalently,

$$T_t = Y^{\text{ref}}\left[-\psi_y(y_t - y^{\text{ref}}) + \nu_t\right]. \tag{42}$$

A high transfer may therefore reflect weak aggregate output or a positive discretionary fiscal state:

$$\widetilde{T}_t \uparrow \quad \Longrightarrow \quad y_t \downarrow \quad \text{or} \quad \nu_t \uparrow. \tag{43}$$

Government debt evolves according to

$$B^g_{t+1} = R^* B^g_t + T_t - \tau_t. \tag{44}$$

Define government debt relative to its reference level as

$$D_t = B^g_t - \bar{B}. \tag{45}$$

Taxes stabilize government debt according to

$$\tau_t = \bar{\tau} + \kappa D_t. \tag{46}$$

At the reference point, $\widetilde{y}_t = 0$, $\nu_t = 0$, and $B^g_t = \bar{B}$, so transfers are zero and

$$\bar{\tau} = (R^* - 1)\bar{B}. \tag{47}$$

Combining the government budget constraint with the fiscal rules gives

$$D_{t+1} = (R^* - \kappa)D_t + \Upsilon^{\text{ref}} \left[-\psi_y (y_t - y^{\text{ref}}) + \nu_t \right]. \tag{48}$$

Because taxes are observed, households infer the current debt deviation exactly:

$$D_t = \frac{\tau_t - \bar{\tau}}{\kappa}. \tag{49}$$

Government debt is therefore an observed aggregate state, rather than an additional hidden signal.

5.3 Information structure and timing

Households observe all variables entering their current budget constraints:

$$\mathcal{I}_{it} \supseteq \{A_{it}, W_{it}, T_t, \tau_t\}. \tag{50}$$

They do not directly observe aggregate output y_t, idiosyncratic efficiency e_{it}, or the discretionary fiscal state ν_t. Aggregate productivity z_t and its innovation μ_t are also unobserved. The two signals about hidden current conditions are

$$w_{it} = y_t + e_{it}, \tag{51}$$

$$\widetilde{T}_t = -\psi_y y_t + \psi_y y^{\text{ref}} + \nu_t. \tag{52}$$

Labor income confounds aggregate output with idiosyncratic efficiency. Transfers confound discretionary fiscal policy with the systematic countercyclical response to aggregate output. Taxes reveal D_t exactly through (49).

The timing within period t is as follows. Aggregate productivity and the fiscal innovation are realized. Given predetermined government debt and the cross-sectional distribution of household states, aggregate consumption, output, transfers, and labor income are determined jointly. Household i observes (W_{it}, T_t, τ_t), updates its beliefs, and chooses consumption and next-period assets.

Absent an approximation, the aggregate state includes productivity, government debt, the fiscal state, and the cross-sectional distribution of household assets, efficiencies, and beliefs. Household beliefs affect saving, saving determines aggregate consumption, aggregate consumption affects output, and output changes the signals households use to form beliefs. The exact forecasting problem is therefore infinite dimensional.

5.4 Perceived law of motion and Bayesian updating

I approximate aggregate log output directly with the perceived law of motion

$$y_t = b_0 + b_y y_{t-1} + b_B D_t + b_\nu \nu_t + b_\mu \mu_t. \tag{53}$$

The productivity innovation enters the perceived law directly, rather than productivity being an additional perceived aggregate state. The law in (53) is a restricted approximation to the simulated equilibrium, not an algebraic consequence of (34).

Because the current transfer and tax payment determine next-period debt exactly, D_{t+1} is known at the end of period t. Under the perceived law, output evolves according to

$$y_{t+1} = b_0 + b_y y_t + b_B D_{t+1} + b_\nu \rho_\nu \nu_t + b_\mu \mu_{t+1} + b_\nu \varepsilon^\nu_{t+1}. \tag{54}$$

For filtering, it is convenient to collect the hidden states in

$$q_{it} = (y_t, e_{it}, \nu_t)'. \tag{55}$$

Define the centered idiosyncratic innovation

$$\varepsilon^e_{i,t+1} = \zeta_{i,t+1} - \mu_\zeta, \qquad \varepsilon^e_{i,t+1} \sim \mathcal{N}(0, \sigma^2_\zeta). \tag{56}$$

The perceived transition can be written as

$$q_{i,t+1} = d_q + A_q q_{it} + D_q D_{t+1} + G_q \varepsilon_{i,t+1}, \tag{57}$$

where

$$d_q = \begin{pmatrix} b_0 \\ \mu_\zeta \\ 0 \end{pmatrix}, \qquad D_q = \begin{pmatrix} b_B \\ 0 \\ 0 \end{pmatrix}. \tag{58}$$

The transition matrix is given by

$$A_q = \begin{pmatrix} b_y & 0 & b_\nu \rho_\nu \\ 0 & \rho_e & 0 \\ 0 & 0 & \rho_\nu \end{pmatrix}, \tag{59}$$

and exogenous aggregate shocks as well as the individual-level shock enter through

$$\varepsilon_{i,t+1} = \begin{pmatrix} \mu_{t+1} \\ \varepsilon^\nu_{t+1} \\ \varepsilon^e_{i,t+1} \end{pmatrix}, \qquad G_q = \begin{pmatrix} b_\mu & b_\nu & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \tag{60}$$

The covariance matrix of the primitive innovations is

$$\Sigma_\varepsilon = \begin{pmatrix} \sigma^2_\mu & 0 & 0 \\ 0 & \sigma^2_\nu & 0 \\ 0 & 0 & \sigma^2_\zeta \end{pmatrix}, \qquad Q_q = G_q \Sigma_\varepsilon G_q'. \tag{61}$$

Define the observation vector as

$$s_{it} = (w_{it}, \widetilde{T}_t)'. \tag{62}$$

The two exact observation equations are

$$s_{it} = h + H q_{it}, \qquad h = \begin{pmatrix} 0 \\ \psi_y y^{\text{ref}} \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 1 & 0 \\ -\psi_y & 0 & 1 \end{pmatrix}. \tag{63}$$

Let

$$m^q_{i,t|t-1} = \mathbb{E}_{it-1}[q_{it}] \tag{64}$$

denote household i's prior mean and let $P_{t|t-1}$ denote the corresponding prior covariance

matrix. Since D_t is observed, the prediction step is

$$m^q_{i,t|t-1} = d_q + A_q m^q_{i,t-1|t-1} + D_q D_t, \tag{65}$$

$$P_{t|t-1} = A_q P_{t-1|t-1} A_q' + Q_q. \tag{66}$$

After observing s_{it}, the Kalman gain is

$$K_t = P_{t|t-1} H' \left(H P_{t|t-1} H' \right)^{-1}. \tag{67}$$

The posterior mean and covariance are

$$m^q_{i,t|t} = m^q_{i,t|t-1} + K_t \left(s_{it} - h - H m^q_{i,t|t-1} \right), \tag{68}$$

$$P_{t|t} = (I - K_t H) P_{t|t-1}. \tag{69}$$

The covariance matrices are common across households. For a given perceived law of motion, they converge to stationary matrices and do not enter the individual household state. The two exact observations for income leave only one independent posterior mean. I use

$$\widehat{y}_{i,t|t} = \mathbb{E}_{it}[y_t] \tag{70}$$

as the reduced belief state. The posterior means of individual log efficiency and the fiscal state are implied by the current signals:

$$\widehat{e}_{i,t|t} = w_{it} - \widehat{y}_{i,t|t}, \tag{71}$$

$$\widehat{\nu}_{i,t|t} = \widetilde{T}_t + \psi_y (\widehat{y}_{i,t|t} - y^{\text{ref}}). \tag{72}$$

One-period-ahead expectations satisfy

$$\mathbb{E}_{it}[\nu_{t+1}] = \rho_\nu \widehat{\nu}_{i,t|t}, \tag{73}$$

$$D_{t+1} = (R^* - \kappa) D_t + T_t, \tag{74}$$

$$\mathbb{E}_{it}[e_{i,t+1}] = \mu_\zeta + \rho_e \widehat{e}_{i,t|t}, \tag{75}$$

$$\mathbb{E}_{it}[y_{t+1}] = b_0 + b_y \widehat{y}_{i,t|t} + b_B D_{t+1} + b_\nu \rho_\nu \widehat{\nu}_{i,t|t}. \tag{76}$$

The deterministic transition for (y_t, D_t, ν_t) implied by the perceived law and fiscal rules

has transition matrix

$$\mathcal{A} = \begin{pmatrix} b_y - b_B\psi_y Y^{\text{ref}} & b_B(R^* - \kappa) & b_B Y^{\text{ref}} + b_\nu \rho_\nu \\ -\psi_y Y^{\text{ref}} & R^* - \kappa & Y^{\text{ref}} \\ 0 & 0 & \rho_\nu \end{pmatrix}. \tag{77}$$

The converged perceived system must satisfy

$$\rho(\mathcal{A}) < 1, \tag{78}$$

where $\rho(\mathcal{A})$ denotes the spectral radius. This joint condition replaces a stability check based only on the partial debt coefficient $R^* - \kappa$.

5.5 Household problem and equilibrium

A convenient post-signal household state is

$$x_{it} = \left(A_{it}, W_{it}, T_t, \tau_t, \widehat{y}_{i,t|t}\right). \tag{79}$$

The remaining posterior means are recovered from current observations using (71) and (72). The household problem can be written recursively as

$$V(x_{it}) = \max_{A_{i,t+1} \geq 0} \left\{ u(C_{it}) + \beta \mathbb{E}_{it} \left[V(x_{i,t+1}) \right] \right\}, \tag{80}$$

where current consumption is

$$C_{it} = R^* A_{it} + W_{it} + T_t - \tau_t - A_{i,t+1}. \tag{81}$$

Future household states are generated by the perceived output law, the fiscal rules, the idiosyncratic efficiency process, and the Kalman update.

For given productivity Z_t, fiscal state ν_t, debt D_t, and the household distribution, aggregate consumption determines output through (34). Output in turn determines individual labor income and the systematic component of transfers. Current consumption, output, transfers, and posterior beliefs must therefore be determined jointly within the period. For a candidate value of C_t, implied output determines the labor-income and transfer signals, households update their beliefs and choose consumption, and the candidate must satisfy

$$C_t = \int_0^1 C\left(A_{it}, W_{it}(C_t), T_t(C_t), \tau_t, \widehat{y}_{i,t|t}(C_t)\right) di. \tag{82}$$

Once this fixed point is found, output, transfers, household choices, and next-period government debt are determined.

5.6 Solution method

The model is solved as a fixed point in the perceived output-law coefficients $(b_0, b_y, b_B, b_\nu, b_\mu)$. Converged coefficients are evaluated in terms of accuracy and compared to alternate signal extraction methods, to assess whether the results approximate a rational expectations equilibrium as intended.

First, I guess the coefficients in (53). Given these coefficients, I construct the perceived state-space system and solve for the stationary Kalman covariance matrices and gain. I then solve the household consumption-saving problem under the perceived aggregate law.

Next, I simulate aggregate productivity, fiscal innovations, idiosyncratic efficiency, government debt, and a large panel of households. In each period, I solve the within-period aggregate-consumption fixed point in (82), then update household assets, efficiencies, and posterior beliefs using the household policy functions and the Kalman recursion. This step is computationally intensive, ensuring markets clear in every period and that the signals observed by households are consistent with the aggregate state of the economy.

Using the simulated time series, I estimate

$$y_t = b_0 + b_y y_{t-1} + b_B D_t + b_\nu \nu_t + b_\mu \mu_t + u_t^y. \tag{83}$$

The estimated coefficients update the perceived law of motion. This outer loop is repeated until the perceived coefficients coincide with those generated by the simulated equilibrium. I verify joint fiscal-output stability using (78). Approximation accuracy is evaluated using the fit of the aggregate output law, iterated forecasts, and household-level signal-extraction checks.

I next study how dispersed information changes the effects of fiscal transfers. I first describe the calibration and the behavior of household beliefs in the simulated equilibrium. I then compare the response to a one-standard-deviation discretionary fiscal innovation under full and dispersed information.

5.7 Calibration

The model is calibrated at a quarterly frequency. Table 4 reports the primitive parameters used for the quantitative results.

Table 4: Calibration, fiscal-policy model

Parameter	Value	Description	Source
Households and idiosyncratic income			
β	0.982	Quarterly discount factor	McKay and Reis (2021)
r^*	0.0074	Quarterly net interest rate	McKay and Reis (2021)
σ	1.5	CRRA coefficient	Attanasio and Weber (1995)
ρ_e	0.863	Idiosyncratic persistence	Guvenen (2009); Guarda (2026)
σ_ζ	0.162	Idiosyncratic innovation s.d.	Guvenen (2009); Guarda (2026)
Aggregate process and demand amplification			
ρ_z	0.95	Aggregate persistence	Boppart et al. (2018)
σ_μ	0.007	Productivity innovation s.d.	Boppart et al. (2018)
ω	0.30	Consumption externality	Krueger et al. (2016)
Fiscal policy			
ψ_y	0.187	Countercyclicality coefficient	Reicher (2011)
κ	0.05	Tax response to government debt	Kindberg-Hanlon (2024)
$\bar{B}$	2.0	Reference government debt	Normalization
ρ_ν	0.90	Fiscal persistence	McKay and Reis (2021)
σ_ν	0.0036	Fiscal innovation s.d.	McKay and Reis (2021); Leeper et al. (2010)

The discount factor and quarterly net interest rate are set to $\beta = 0.982$ and $r^* = 0.0074$. The resulting inequality $\beta(1 + r^*) < 1$ ensures that the incomplete-markets economy does not generate unbounded asset accumulation. The CRRA coefficient is $\sigma = 1.5$, and households cannot borrow.

The idiosyncratic-income process uses $\rho_e = 0.863$ and $\sigma_\zeta = 0.162$, based on the quarterly mapping of the earnings process estimated by Guvenen (2009), where the parameters follow Guarda (2026) in their mapping to the simple AR(1) idiosyncratic efficiency process of this paper. Aggregate productivity has persistence $\rho_z = 0.95$ and innovation standard deviation $\sigma_\mu = 0.007$. The demand-amplification coefficient is $\omega = 0.30$.

Fiscal policy has countercyclicality coefficient $\psi_y = 0.187$, following Reicher (2011). The discretionary fiscal state has persistence $\rho_\nu = 0.90$ and innovation standard deviation $\sigma_\nu = 0.0036$, set following McKay and Reis (2021), where the latter is obtained by scaling

their parameter using the roughly 14% transfers-to-GDP ratio from Leeper et al. (2010). Taxes respond to debt with $\kappa = 0.05$, and reference debt is normalized to $\bar{B} = 2.0$. This is chosen such that, in the solved model it implies an annual debt-to-GDP ratio around 50%.

5.8 Aggregate approximation and household beliefs

Solving the model yields the perceived output law reported in Table 5. The aggregate approximation is very accurate, with $R^2 = 0.9999897$.

Table 5: Estimated output law of motion

	b_0	b_y	b_B	b_ν	b_μ	R^2
Coefficient	1.385×10^{-5}	0.960833	-5.24010×10^{-4}	0.00853032	1.10480	0.9999897

Figure 5 compares simulated output with the path implied by the converged perceived law. The two series are nearly indistinguishable. The low-dimensional law therefore captures the aggregate dynamics generated by the full distribution of household states and beliefs.

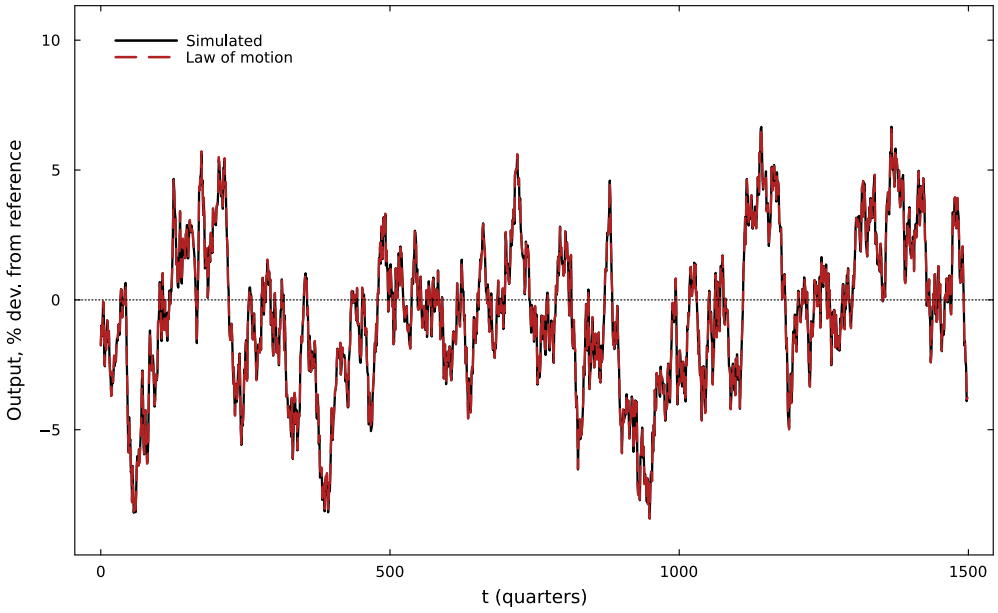


Figure 5: Simulated aggregate output and the perceived law of motion

This high aggregate fit does not imply that households know the current aggregate state. Households observe the signals generated by the economy, not realized aggregate output itself. Figure 6 compares true output with household output beliefs. Plotted is the cross-sectional average belief at each point in time, which tracks true output. Dispersed

information leads to underreaction of beliefs to movements in aggregate output. Additionally, individual posteriors remain dispersed because labor income combines aggregate output with idiosyncratic efficiency, while transfers combine the countercyclical policy response with the discretionary fiscal state. The shaded area in 6 indicates the posterior standard deviation of an agent with the exact mean belief at any given point in time.[3]

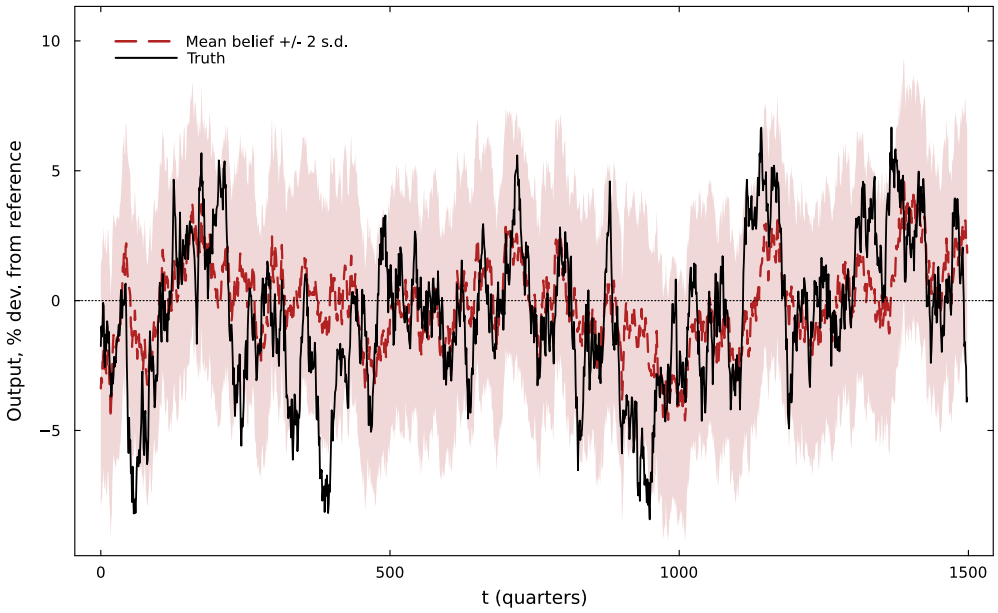


Figure 6: Aggregate output and the distribution of household beliefs

5.8.1 Ex-post signal-extraction checks

The perceived law of motion approximates aggregate output extremely closely, but it remains possible that the Kalman filter imposes an overly restrictive mapping from household observations to output beliefs. I assess this possibility using ex-post machine-learning counterfactuals. These exercises hold the simulated equilibrium fixed and replace the model's filtering rule with alternative forecasting methods. They do not re-solve the household problem or recompute the equilibrium under the alternative beliefs.

The alternatives receive the same current observables as households in the model:

$$o_{it} = (w_{it}, \widetilde{T}_t, D_t)'. \tag{84}$$

I consider a ridge regression and a k-nearest-neighbor estimator using twelve quarters of

[3]Note that this individual posterior variance is always higher than the cross-sectional variance of posterior estimates $\mathrm{Var}_i(\hat{y}_{it|t})$, see e.g. Lahiri and Sheng (2010).

observations, as well as a gated recurrent unit that summarizes the history recursively. Each method is trained to forecast current aggregate output y_t. I simulate 1,000 periods for 100 households and repeat the exercise 10 times.

Table 6 reports out-of-sample forecast performance. To compare with the implementation, the baseline is listed as storing the three posterior means associated with (y_t, e_{it}, ν_t). The final column reports the share of evaluated household-period forecasts for which the alternative produces a larger forecast error than the model belief.

Table 6: Ex-post recovery of aggregate output from household observations

Belief rule	Stored states per household	R^2	Larger error than model belief
Model baseline	3	0.281	N/A
GRU ($H = 16$)	16	0.232	80%
Ridge ($L = 12$)	33	0.237	80%
kNN ($L = 12, k = 160$)	33	0.194	88%

None of the alternative forecasting methods improves on the model belief. The baseline filter achieves an R^2 of 0.281, compared with 0.237 for Ridge, 0.232 for the GRU, and 0.194 for kNN. The alternatives also produce larger household-level forecast errors in at least 80% of the evaluated observations.

This result is consistent with the near-exact fit of the perceived aggregate law. *If* the perceived law of motion were exact, the linear-Gaussian structure would make the Kalman posterior mean the minimum-mean-squared-error forecast conditional on household observations. The ex-post exercise shows that the small approximation error in the perceived law is not large enough to allow different signal extraction frameworks to confer better information.

These alternatives are also infeasible as equilibrium belief rules in the current solution method. Even the smallest selected GRU carries a sixteen-dimensional recursive state. Under the discretization used to solve the household problem, incorporating this state would expand the computational burden by more than a factor of 10^8. Appendix C provides the construction and details for the three counterfactual forecasting rules.

6 Quantitative results

Before turning to fiscal-policy responses, I compare the calibrated model with several moments not targeted in the calibration. Table 7 reports measures of household wealth,

consumption responses, and beliefs about aggregate output.

Table 7: Untargeted model moments

Moment	Full information	Dispersed information	Data
SD of current-output beliefs	0	0.195	–
SD of GDP-growth backcasts	0	0.032	≈ 0.27
GDP-belief/income-change slope	0	0.120	$0.067 - 0.294$
Liquid wealth / annual income	0.56	0.57	0.56
Average quarterly cash MPC	0.119	0.116	0.25

Notes: Belief moments are in log percentage points. The income slope relates GDP-growth backcasts to six-month changes in log household income. The table reports the empirical 95% confidence interval, with 0.181 being the point estimate. The empirical backcast SD is inferred from categorical responses using an interval-normal model and may contain reporting variation. No direct empirical counterpart is available for disagreement about current output levels. The MPC benchmark includes nondurables and services, excluding the durable expenditures included in Appendix A.1.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, BOP-HH wave 68, own calculations (belief moments); Kaplan and Violante (2022) (liquid wealth); Kueng (2018) (MPC). Model calculations by author.

The model generates liquid wealth close to the empirical benchmark, while its average cash MPC falls below the estimate from Kueng (2018). This does not come as a surprise: As emphasized by Kaplan and Violante (2022), relatively low MPCs are a common feature of one-asset incomplete-markets models, which face a trade-off between generating large spending responses and matching realistic household wealth holdings. Overall, neither moment differs substantially between full and dispersed information.

Turning to beliefs, dispersed information generates disagreement about aggregate conditions that is absent under full information. In the model, disagreement is concentrated in beliefs about the current level of output, whereas backcasts of quarterly growth are considerably less dispersed. This is because differences in estimated output levels largely cancel when taking changes between adjacent quarters. The BOP-HH survey elicits the latter, although it is unclear how closely respondents' assessments of past growth correspond to this model object. The estimated survey dispersion exceeds the model's backcast disagreement, but may also reflect measurement and reporting variation. More directly related to the information mechanism, the model reproduces the positive association between household income changes and output beliefs observed in the data, which is absent under full information. Appendix B.8 details the construction of the survey and model belief moments.

Section 6.1 next evaluates the information channel more directly using the randomized fiscal-policy information experiment.

6.1 Validation against the fiscal-policy information experiment

I next map the BOP-HH expectations into the quantitative model. Wave 4 of the data is particularly useful, because it contains the information treatment described in section 3.2. First, all households in this wave are asked about GDP expectations going forward. The data can thus be used to observe how much households' expectations adjust following a large output decrease. Second, the information treatment can be used to directly measure my proposed information channel: Households are informed about a fiscal expansion, and their expectations and spending responses can be compared to the quantitative model.

I set Y^{ref} equal to Q1 2020 output and use a negative shock to aggregate productivity z_t to reproduce the observed decline in German output between Q1 and Q2 2020, a contraction of roughly 9%. Figure 7 shows this decline, the subsequent realized output path, and the model's conditional output distribution following the announcement. The survey was conducted in April 2020. In the quarterly model, the realized Q2 contraction calibrates the economic environment at which households form their forecasts. Households under dispersed information continue to infer aggregate conditions from their usual observations of labor income, transfers, and taxes, whereas households under full information observe the current aggregate state.

First, consider the untreated households in the data. I map these into the dispersed information model as households that have observed their individual cash-flows, but not aggregate conditions. Because their signals are noisy, households' expectations do not fully adjust. On average, the posterior estimate about current output will then sit above the realized aggregate -9% relative to the reference level. In the full information economy, households instead perfectly observe the productivity shock and resulting output movement. From these posterior estimates, forecasts about output going forward can then be computed. The questionnaire elicits expectations about GDP changes over the next twelve months. The model counterpart is expected average output over Q2 2020 through Q1 2021, expressed relative to Q1 2020 output. The resulting data and model results are displayed in figure 8.

Whereas full information implies full adjustment, vastly overshooting the empirical belief response, the untreated dispersed-information forecast places average output over the four-quarter window approximately 1.9% below Q1 2020 output, close to the empirical

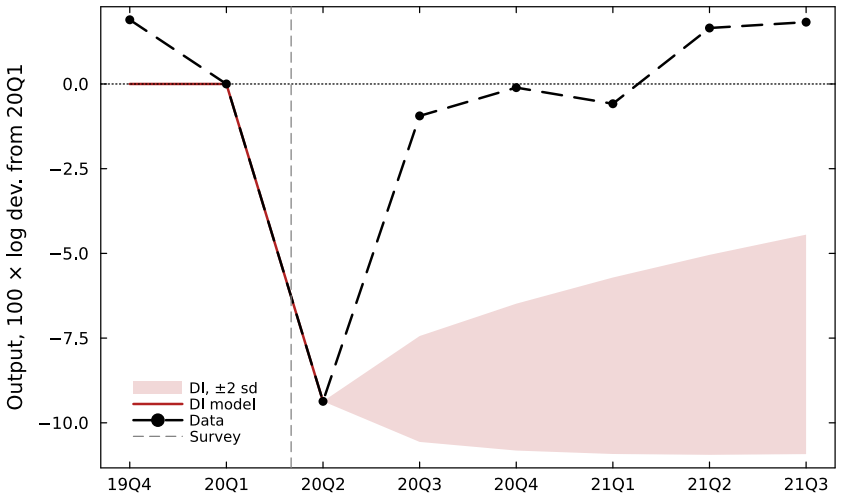


Figure 7: Output around the fiscal-policy information experiment

Notes: German real GDP and the dispersed-information model's conditional output distribution, relative to Q1 2020. The vertical line marks the April 2020 survey; shading covers two conditional standard deviations on either side of the mean following the announcement.

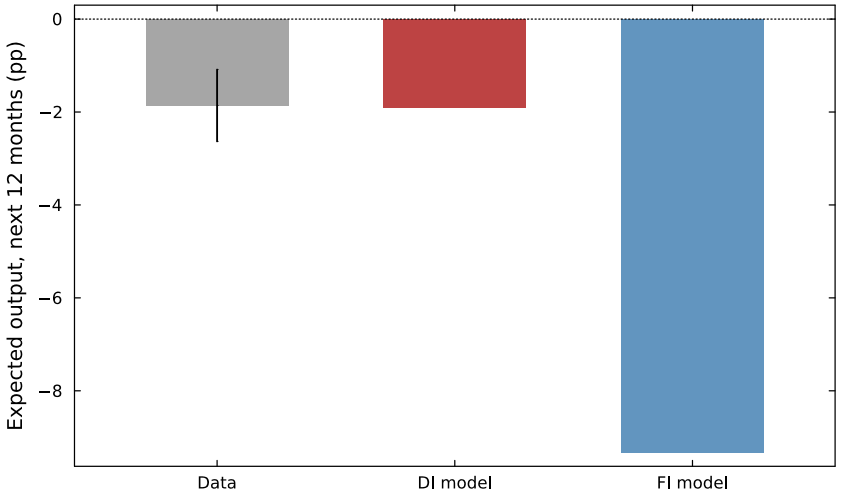


Figure 8: GDP expectations, untreated households

Notes: Data shows the covariate-adjusted placebo forecast from the survey-weighted regression in Table 3, averaged over the estimation sample's covariate distribution. The dispersed information (DI) model shows an average across households that have observed their current cash-flows. Full information (FI) expectations are uniform across households.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations. Model calculations by author.

GDP-growth forecast of -1.86%.[4] Despite the large realized contraction, households' usual cash-flow observations only partially reveal the deterioration in aggregate conditions. Their forecasts therefore adjust much less than those of full-information households, who observe the current aggregate state and anticipate that the contraction will persist. The model thus reproduces the relatively modest pessimism of untreated respondents following a large aggregate downturn.

Next, I make use of the information treatment from the data. In the model, the treatment is given to a measure zero of households. It thus does not affect current aggregates. Treated households do however fully anticipate future general-equilibrium feedback the transfers will induce once they are paid out. Within each information structure, I compare households that retain their usual information set with households that additionally receive the fiscal announcement.

The treatment describes a EUR 750 billion federal fiscal package, of which EUR 600 billion is support for firms (Deutsche Bundesbank, 2020). I map the remaining EUR 150 billion to the model's cumulative household-transfer component over four quarters, from Q2 2020 through Q1 2021. Using EUR 884 billion as the Q1 2020 reference value of quarterly output, the normalized announced amount is

$$s = \sum_{j=0}^{3} \widetilde{T}_{t+j} = \frac{150}{884} = 0.1696. \tag{85}$$

Households have already observed the current transfer. The announcement commits then the government to a fiscal path. Households expect the plan to be carried out over the next quarters, but realized transfers still respond to future productivity through the transfer rule, so the realized total equals s only in expectation. The fiscal path does not specify the quarterly allocation of the package over future periods. Households therefore infer how much of the announcement reflects previously unobserved current economic conditions and how much reflects planned discretionary fiscal interventions over the following three quarters.

The government is assumed to know the current economic state and its own fiscal plan, but not future productivity innovations. Consequently, the announcement contains no additional information about future productivity shocks. It provides one signal about two conceptually distinct objects: the current hidden state, including output, and the sequence of planned fiscal innovations. Under full information, it would instead exclusively inform

[4] This is the survey-weighted, covariate-adjusted placebo forecast, which is close to the mean across all untreated households of -1.73%

households about planned fiscal innovations, with the current aggregate state already perfectly observed.

Households allocate the announcement surprise between these explanations according to their prior uncertainty. The inferred plan is front-loaded because an innovation in the first future quarter affects transfers in all three remaining quarters, whereas a later innovation affects fewer quarters. The announcement fixes the cumulative transfer amount, but individual quarterly transfers remain uncertain. Appendix D provides the exact signal-extraction problem and the numerical decomposition of the announcement surprise.

Learning about the current state is central for GDP expectations because a downward revision in current output persists throughout the four-quarter forecast window. Figure 9 compares treatment effect from the data with the effect of the policy announcement in the dispersed information and full information model.

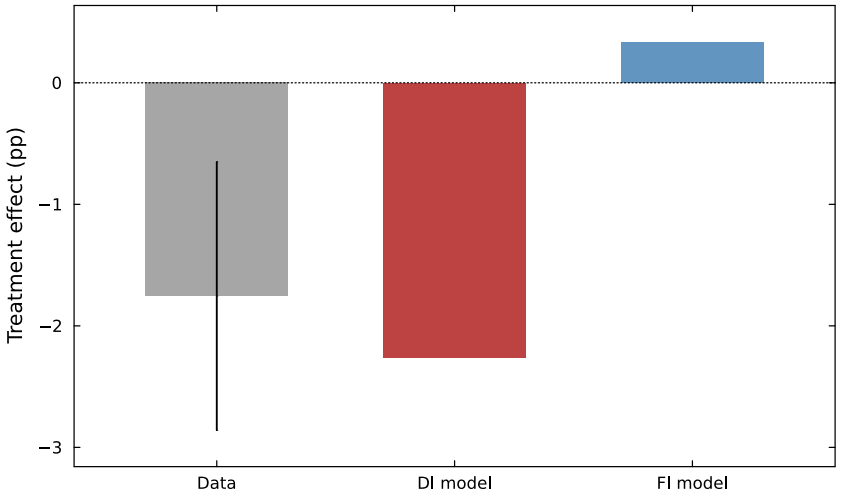


Figure 9: GDP expectations, treatment effect

Notes: The treatment provides information about Germany's EUR 750 billion fiscal package; the model maps EUR 150 billion to household transfers over four quarters. Plotted is the treated-minus-untreated difference in percentage points.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations. Model calculations by author.

The information treatment, or policy announcement, lowers expected output by approximately 2.3 percentage points in the model, compared with 1.76 percentage points in the data. Under full information, households already know the current state and interpret the announcement only as news about the fiscal plan. The resulting treatment effect is positive, at approximately 0.34 percentage points. The negative empirical response therefore requires households to learn something about current aggregate conditions from the

announcement.

Household-income expectations provide a complementary check on how the announcement is mapped into expected resources. The survey elicits a probability distribution over changes in monthly net household income over the following twelve months, including after-tax labor income, pensions, and transfers. Panel A of Table 8 compares the treatment effect in the data and model. The empirical estimate is a positive but imprecise 2.21%, compared with 2.98% under dispersed information and 3.39% under full information. Both model predictions are consistent with this evidence, so the result does not distinguish between the information structures. In the dispersed-information model, higher expected transfers more than offset lower expected labor income. Pessimism about aggregate conditions can therefore coexist with higher expected household income. Appendix D.7 reports the income decomposition and heterogeneity estimates.

To compare spending magnitudes, I translate the empirical purchase-probability responses into consumption-equivalent revisions, referred to below as consumption revisions. First, control-group purchase probabilities and baseline expenditure on cars and other durables convert the probability responses into changes in expected durable expenditure, holding expenditure conditional on purchase fixed. The categories are weighted by baseline net expenditure, using the German EVS 2018 expenditure data reported by Statistisches Bundesamt (Destatis, 2020) and deducting resale proceeds for cars. Second, following the durable-stock mapping in Laibson et al. (2022, Appendix B), I assume that desired durable stocks move proportionally with consumption and use quarterly depreciation of 5% to translate durable expenditure into a consumption revision. Appendix D.6 gives the derivation and calibration; the raw purchase-probability estimates are reported in Appendix B.4.2.

Panel B of Table 8 compares these consumption revisions with the model. The empirical point estimate is -2.54%, while the dispersed-information model produces a smaller decline of 0.31%. The full-information model instead predicts an increase of 1.43%. The approximate empirical 95% confidence interval includes the dispersed-information prediction and zero, but excludes the full-information prediction. The comparison therefore supports the negative response generated by dispersed information, although the empirical magnitude is imprecisely estimated.

Dispersed information also reproduces the negative income and prior-belief gradients: consumption revisions become more negative among households with higher incomes or more optimistic priors. Both empirical gradients are imprecisely estimated. The negative income gradient is present under both information structures and therefore does not by itself identify the information channel. By contrast, heterogeneity by prior beliefs is

Table 8: Income expectations and consumption revisions in the experiment and model

	Data	Dispersed information	Full information
Panel A: Household-income expectations			
Expected net-income revision	2.21	2.98	3.39
	(2.73)		
Panel B: Consumption revisions			
Average consumption revision	−2.54	−0.31	1.43
	(1.54)		
Slope across income brackets	−1.83	−0.53	−0.40
	(1.30)		
Slope across prior categories	−2.42	−0.60	–
	(1.66)		

Notes: Entries are treated-minus-untreated revisions in expected household net income and consumption, in percent. Empirical consumption revisions are consumption-equivalent measures obtained from purchase intentions using the mapping in Appendix D.6; model entries are average consumption revisions over four quarters. Slope rows report the change in the consumption revision, in percentage points, per one-category increase in household income or prior optimism about growth. Empirical standard errors are in parentheses.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations. Expenditure inputs: EVS 2018, Destatis (2020). Model calculations by author.

inherently linked to dispersed information.

I do not use the survey's savings-related outcomes as quantitative validation targets. The estimated treatment effect on the probability of building additional financial reserves is negative but statistically insignificant, whereas respondents in the following survey wave report lower probabilities of difficulty covering expenditure. Building reserves may be interpreted as longer-term accumulation, while the follow-up questions concern near-term liquidity and the ability to finance expenses from income or available funds. These outcomes therefore do not provide a clean counterpart to model saving. Appendix B.7 reports the estimates and question wording.

Taken together, the exercise evaluates the model against aggregate expectations, household-income expectations, and consumption revisions. The dispersed-information model approximately reproduces the level and negative treatment effect of GDP expectations, while both information structures yield household-income revisions consistent with the imprecise empirical estimate. For consumption, dispersed information reproduces the negative average response and the heterogeneity gradients, with a smaller decline than the empirical point estimate. The next subsection returns to the model's aggregate consumption

response to a discretionary fiscal intervention.

6.2 Main results, fiscal policy

6.2.1 Aggregate consumption response

I next use the model to study a one-standard-deviation positive innovation to the discretionary fiscal state, $\varepsilon_t^{\nu} = \sigma_{\nu}$. Because ν_t is persistent, the innovation raises both current transfers and expected future transfers. The shock is evaluated from the resting distribution under full and dispersed information.

Figure 10 compares the aggregate consumption responses directly. Under full information, the impact consumption response, normalized by the fiscal payment, is about 0.38. Under dispersed information, the corresponding response is approximately 0.07 and the consumption adjustment is substantially more gradual, as households learn about the discretionary component of transfers over time. As households increasingly attribute transfers to this component, the adverse information channel weakens, allowing consumption to rise and generating the hump-shaped aggregate response. On balance, the fiscal payment is much less effective at raising near-term aggregate consumption when households must infer why it was paid.

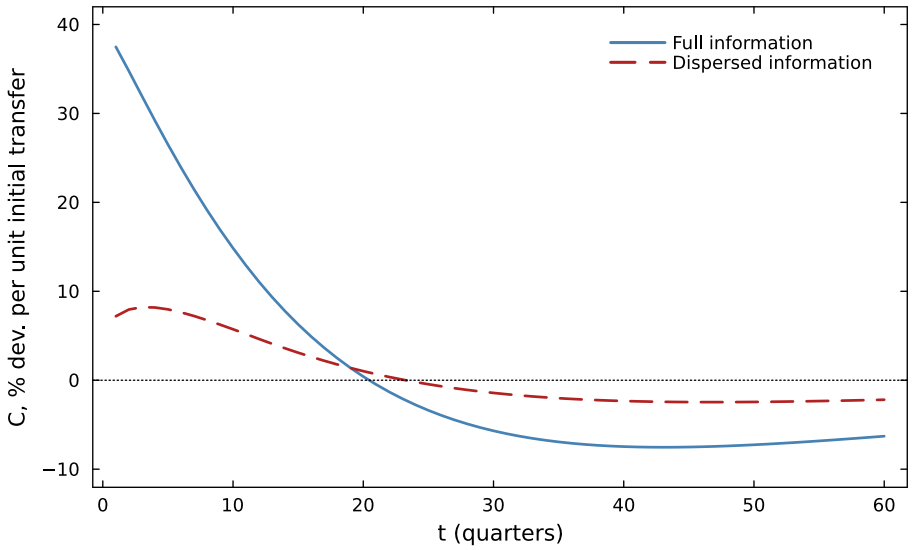


Figure 10: Aggregate consumption response under full and dispersed information

Notes: A one-standard-deviation discretionary fiscal innovation raises current and expected future transfers. Consumption responses are normalized by the initial transfer and multiplied by 100, so a value of 38 denotes 38 cents per dollar; periods are quarters.

Note that, in this model, the aggregate consumption response follows from an aggregation over all micro responses. Empirical estimates of spending responses to payments can then serve as a benchmark for the response, allowing for a comparison between the model and empirical evidence. Appendix A.1 describes these studies in more detail.

Empirical estimates of household spending responses to cash receipts, such as Souleles (1999) and Kueng (2018), are generally large. For countercyclical payments, the empirical evidence is mixed. Orchard et al. (2026) find a negative quarterly MPC of -0.07 out of the 2001 rebate payments, although the estimate is imprecise. For the 2008 stimulus episodes, Orchard et al. (2025) find an MPC of 0.28, although the estimated response excluding cars is -0.02. For transfers during the COVID-19 pandemic, Parker et al. (2022) find an unweighted average MPC of 0.16 across the three EIP rounds. The response is especially low in the third round of payments, with an MPC of 0.015. Goldfayn-Frank et al. (2022) find a similarly low MPC of 0.054 from payments in Germany during the same time period, although this is a monthly estimate.

The studies mentioned cover several (near) zero MPC estimates from countercyclical transfers. Compared with the 0.25 MPC (0.38 when including durables) estimate from Kueng (2018), these observations suggest a gap between the spending response to some pro- or acyclical payments and the much weaker response observed in countercyclical episodes. The full information model has consumption move 0.38 in response to a fiscal expansion, although this is not a pure MPC as it contains an (anticipated) policy component as well. The pure cash MPC in the model is about 0.12. Turning to the consumption responses to countercyclical payments, I consider a zero spending response as a benchmark for these low responses. In terms of explaining why a countercyclical payment may instead imply a (near) zero spending response, the dispersed information model closes 80% of the gap between the full information response and this benchmark, with the impact response of 0.07 much closer to the zero benchmark.

6.2.2 Consumption responses across the household distribution

The experiment is a positive discretionary fiscal innovation, which raises transfers to all households. Under dispersed information, households observe the payment but cannot perfectly distinguish a discretionary increase from the policy's countercyclical response to weak aggregate output.

Figure 11 reports the impact consumption response by pre-shock cash-on-hand decile under full information. Responses are divided by the size of the current transfer and can be interpreted as impact consumption responses per unit of fiscal payment.

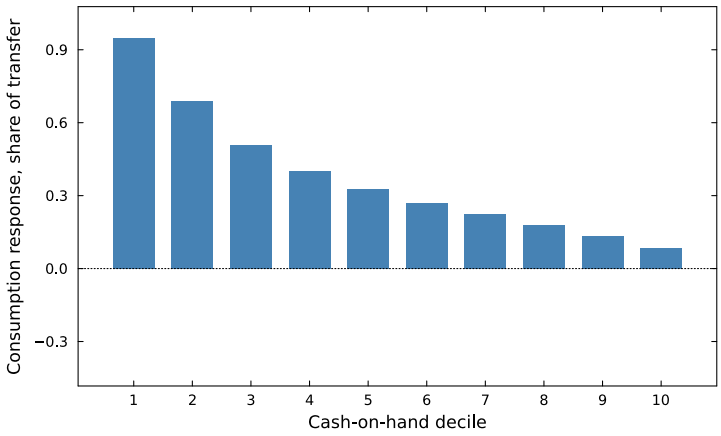


Figure 11: Impact consumption response under full information

Notes: Mean impact consumption responses per dollar transferred after a one-standard-deviation discretionary fiscal innovation, by pre-shock cash-on-hand decile (1 = lowest; 10 = highest). Under full information, households observe aggregate conditions and recognize the discretionary innovation.

Under full information, the impact response has the standard incomplete-markets shape. It is largest among households with low cash on hand and declines along the distribution. Households know that the payment reflects a persistent discretionary fiscal innovation and therefore anticipate additional future transfers, as well as the associated indirect general-equilibrium feedback. Anticipated future taxes are also part of this policy component, and decrease the response, but are outweighed by the aforementioned forces. Under full information, there are thus two effects of the payment. First, a cash effect, with a transfer of size T raising a household's current resources directly, holding expectations fixed. Second, the anticipated effects of the policy, including future transfers, taxes, and changes in household income driven by general-equilibrium feedback.

Figure 12 reports the corresponding response under dispersed information. While responses are now generally lower across the distribution, the change is especially striking towards the top: Households with substantial cash on hand actually lower their consumption upon this increase in resources. This is the model's dispersed information channel at work.

The direct cash-flow effect, as well as the anticipated policy effect, is positive under both information structures. The cash effect, essentially the pure MPC, of the transfer is still positive, with an increase in resources raising a household's consumption. Similarly, the policy component is positive. Even though households can no longer identify perfectly that transfers are discretionary, the policy effect remains large. As under full information, the expected general-equilibrium effects and future transfers outweigh the effect of expected

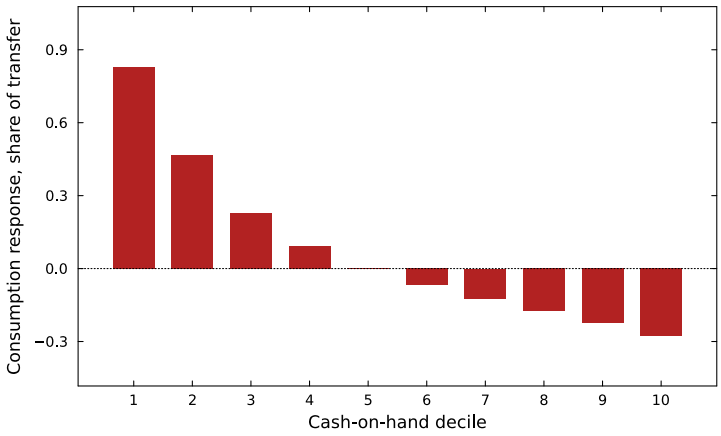


Figure 12: Impact consumption response under dispersed information

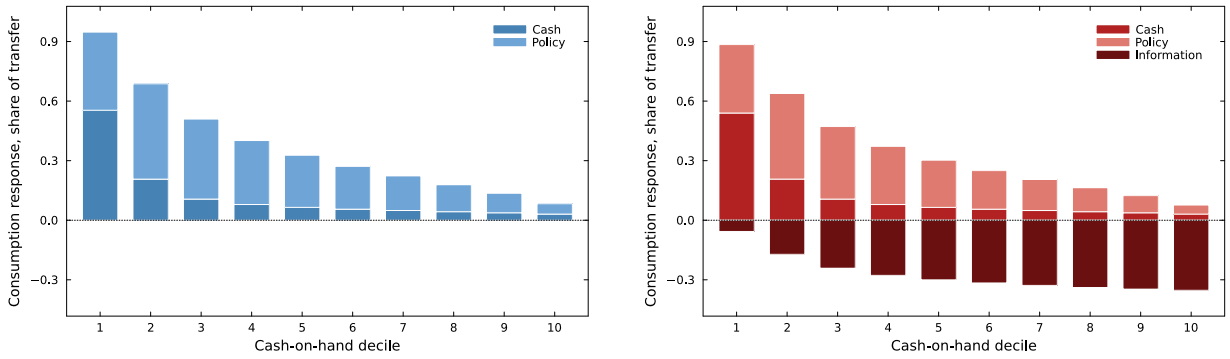
Notes: Mean impact consumption responses per dollar transferred after a one-standard-deviation discretionary fiscal innovation, by pre-shock cash-on-hand decile (1 = lowest; 10 = highest). Households observe the payment but cannot fully distinguish discretionary from countercyclical stimulus.

future taxes. In contrast to the full-information economy, the information channel proposed by this paper now arises. Households do not observe aggregate output directly, and infer from a transfer receipt that it may be low. This lowers expectations about future output and household income, lowering consumption.

Figure 13 decomposes the impact response into three components. The cash effect and policy effects are defined as before. The information channel captures the additional response arising because households use transfer receipts to infer aggregate output. This channel therefore separates what the payment reveals about aggregate conditions today from the anticipated effects of the policy itself. In the figure, these components are labelled Cash, Policy, and Information.

Despite the two economies differing in the degree of risk households are exposed to, and thus in their wealth distributions, the direct cash effect and indirect policy effects are very close across the two information structures. The difference stems almost entirely from the information channel: Under full information, there is no additional inference from a transfer payment about aggregate conditions, so the cash and policy components account for the entire response.

Under dispersed information, the information channel is negative throughout the distribution and becomes increasingly negative higher in the distribution. Households with sufficiently high cash on hand respond strongly to the negative revision in expected aggregate output. For these households, the information channel more than offsets the positive cash and anticipated policy effects, reversing the sign of the total response. It is also the



(a) Full information

(b) Dispersed information

Figure 13: Cash, policy, and information components of the impact consumption response.

Notes: Impact responses per dollar transferred after a discretionary fiscal innovation, by pre-shock cash-on-hand decile. Cash captures additional current resources; Policy captures anticipated transfers, taxes, and general-equilibrium effects; Information captures spending changes arising from what the payment reveals about current aggregate conditions.

information component that induces the hump-shaped aggregate response, detailed above. The channel is negative because households infer that fiscal transfers may be high because aggregate conditions are weak, even if the transfers are truly discretionary. Over time, households update their expectations, and the discretionary nature of transfers becomes more apparent. This increases expectations about aggregate output, and thus household income, leading to higher consumption.

The increasing importance of the information channel across the distribution does not reflect larger downward revisions in output expectations among households with more cash on hand. Figure 14 compares average beliefs about current aggregate output with and without the fiscal innovation, by cash-on-hand decile. Households toward the top of the distribution are generally more optimistic about current aggregate output. This relationship is driven by wages: Positive wage shocks raise current resources and, because wages are a signal of aggregate output, also imply more optimistic beliefs about aggregate conditions.

Following the fiscal innovation, output beliefs fall by a similar amount across deciles. The stronger negative consumption response at the top of the distribution therefore follows from differences in spending responses to deteriorating expectations, rather than larger belief revisions. Households closer to the borrowing constraint are less responsive to the implied decrease in future income, while the additional current resources provided by the transfer have large effects on their consumption.

Taken together, the results show that the information channel of fiscal policy materially

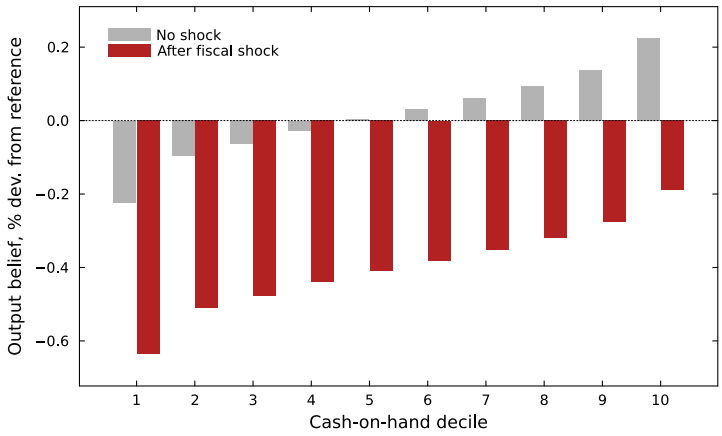


Figure 14: Beliefs about current aggregate output by cash-on-hand decile

Notes: Mean beliefs about current aggregate output with and without a discretionary fiscal innovation under dispersed information. Households are grouped by pre-shock cash-on-hand decile (1 = lowest; 10 = highest); beliefs are expressed as percentage deviations from reference output.

changes both the distributional and aggregate effects of transfers. The direct cash-flow and policy responses remain similar across the two structures. What changes is the interpretation of the policy. Because a transfer can signal weak aggregate conditions, the information channel reduces households' consumption, especially among rich households with substantial scope to adjust.

7 Extension: Targeted transfers

The results above suggest that targeting transfers toward households at the bottom of the distribution may increase the effectiveness of fiscal policy through both resources and information. Households with few resources respond to additional cash flows with larger consumption increases, while households with substantial cash on hand respond to the information channel with larger decreases. Targeting can thus direct resources toward households with high spending responses while decreasing the impact of the information channel among households whose consumption is particularly sensitive to expectations.

I compare universal transfers with two settings in which payments are directed toward the bottom half of the contemporaneous income distribution. A simplified exercise retains the baseline household policies and perceived law of motion, while allowing output to adjust endogenously. The full general-equilibrium model additionally has households incorporate the targeted transfer rule into their decisions and expectations. This implies that households learn about the aggregate state not only from observed transfers, but also

from whether they are targeted at all: Households observe their own wage directly, and learn from their transfer eligibility whether their wage sits below or above the median. Because this introduces a non-Gaussian signal, the solution approach is extended to incorporate this.

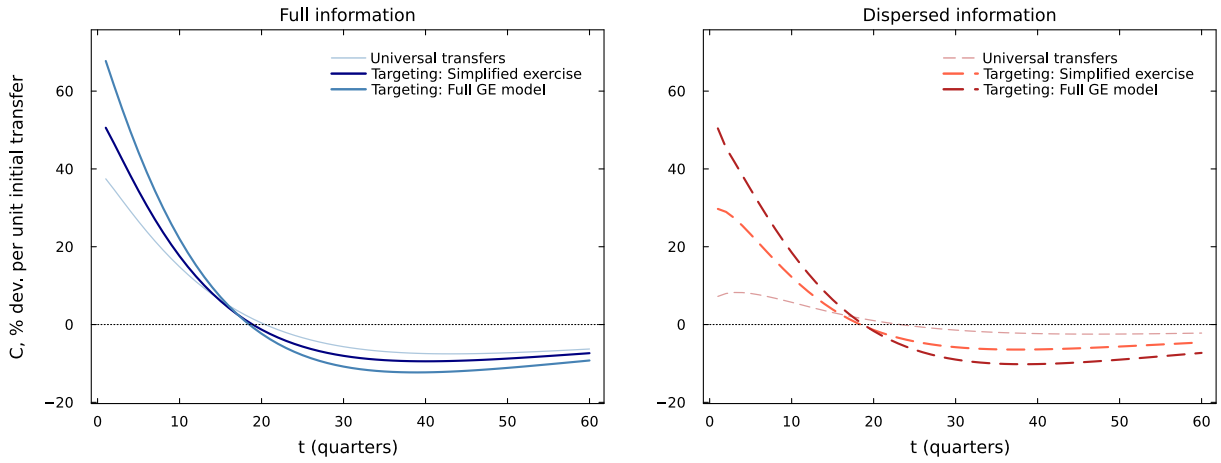


Figure 15: Aggregate consumption responses under universal and targeted transfers

Notes: Responses to the same discretionary fiscal innovation, normalized by aggregate transfers at impact.

Figure 15 compares the aggregate response paths across the three settings. Targeting increases the initial consumption response under both information structures, with a larger gain under dispersed information. It also makes the response front-loaded rather than hump-shaped. Solving the full model raises the response further relative to the simplified exercise. Dispersed information continues to reduce the initial response, but the difference from full information is much less than under universal transfers.

I first examine the simplified exercise to explain why reallocating transfers is particularly effective under dispersed information. I then describe the full targeted model, the method used to solve it, and examine the additional increase in consumption when households incorporate the targeted rule into their decisions.

7.1 Results: Simplified targeting exercise

I reallocate transfers toward the bottom half of the current income distribution. Since individual labor income satisfies $W_{it} = Y_t E_{it}$, households' income ranks coincide with their ranks in idiosyncratic efficiency. Let $\bar{e} = -\sigma_e^2/2$ denote the median of log efficiency.

Eligibility and individual transfers are

$$d_{it} = \mathbf{1}\{e_{it} \leq \bar{e}\}, \tag{86}$$

$$T_{it} = 2T_t d_{it}. \tag{87}$$

Half of the population receives twice the universal payment, while the remaining households receive no transfer. At any given aggregate state, total payments remain T_t. The aggregate fiscal rule, the discretionary fiscal innovation, and the tax rule are the same as in the universal-transfer economy.

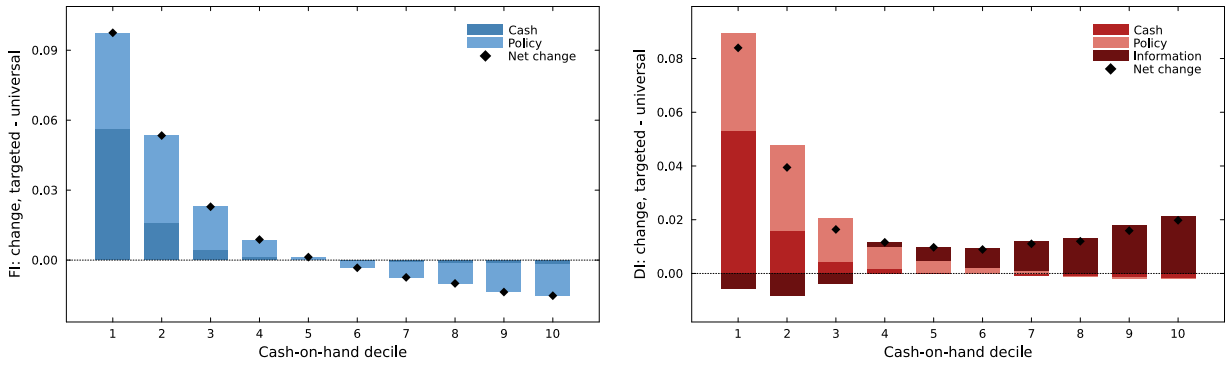
The exercise retains the baseline consumption functions, perceived output law, and initial household distribution for each information structure. Output remains endogenous: Changes in consumption feed back into output and individual labor income through the consumption externality. The exercise therefore includes realized general-equilibrium feedback, but does not re-solve household decisions or the perceived law of motion under a targeted rule.

Reallocation raises the impact consumption response from approximately 0.38 to 0.51 per dollar under full information, and from approximately 0.07 to 0.30 under dispersed information. Targeting thus produces a larger absolute gain under dispersed information, increasing the response by about 0.23 rather than 0.13.

Figure 16 decomposes these gains by pre-shock cash-on-hand decile. These bars report each decile's contribution to the change in the aggregate response, and thus do not measure a response per dollar received. Eligibility is based on current income. The horizontal axis groups households by cash on hand, which includes wealth. Recipients can therefore appear throughout the cash-on-hand distribution.

Under full information, the gains are concentrated among households with little cash on hand. Reallocating resources toward these households increases spending because their consumption responds strongly to current resources. Contributions decline toward the top of the distribution and eventually become negative as households receive fewer transfers than in the universal regime.

Under dispersed information, the net change is positive throughout the cash-on-hand distribution. At the bottom, additional cash and policy effects raise consumption. Toward the top, the improvement instead comes mainly from the information channel: These households receive fewer transfers, but are also less exposed to the negative signal about aggregate conditions conveyed by the payments. The reduction in the information channel more than offsets the decrease in the cash and policy components for these households. This reduction also makes the aggregate response front-loaded. Targeting shifts payments



(a) Full information

(b) Dispersed information

Figure 16: Contributions to the change in the consumption response from reallocation

Notes: Differences between targeted and universal transfers, decomposed into cash, policy, and information components. Bars report contributions by pre-shock cash-on-hand decile to the change in the aggregate consumption response per dollar of aggregate transfers.

toward households whose spending responds strongly to current resources and less strongly to expectations, while reducing the adverse information effect among households most sensitive to news. The immediate spending response therefore dominates the later adjustment associated with learning.

Targeting therefore operates through two margins. It directs payments toward households whose spending responds strongly to resources, as in the full-information economy. In addition, it lessens the information channel among households whose consumption responds most strongly to deteriorating expectations. The latter margin accounts for the larger gain under dispersed information.

7.2 General equilibrium with targeted transfers

I next solve the model with the targeted rule in equations (86) and (87). Eligibility is reassessed every quarter, and households understand that their future payments depend on their future income ranks. Household consumption functions and the perceived aggregate law are solved jointly in general equilibrium under this rule.

Targeting changes the information available to households. Recipients observe the payment amount and can recover aggregate transfers as $T_t = T_{it}/2$. They also learn that their idiosyncratic efficiency lies below the eligibility cutoff. Nonrecipients do not observe the current aggregate payment, but learn that their efficiency lies above the cutoff.

Since households observe their own labor income, eligibility also contains information

about aggregate output. Receiving a transfer implies $e_{it} \leq \bar{e}$, and therefore $y_t \geq w_{it} - \bar{e}$. Nonreceipt implies the opposite inequality. Let $\pi^G_{R,it}$ and $\pi^G_{N,it}$ denote the Gaussian distributions obtained after processing the continuous observations available to recipients and nonrecipients, respectively. Incorporating eligibility gives

$$\pi_{it}(dq) \propto \begin{cases} \mathbf{1}\{y_t \geq w_{it} - \bar{e}\}\, \pi^G_{R,it}(dq), & \text{if recipient,} \\ \mathbf{1}\{y_t < w_{it} - \bar{e}\}\, \pi^G_{N,it}(dq), & \text{if nonrecipient.} \end{cases} \tag{88}$$

Eligibility thus removes output values inconsistent with the household's observed income and transfer status. Conditional on the continuous observations, eligibility shifts output beliefs upward, while noneligibility shifts them downward. This information is distinct from the adverse signal about aggregate conditions that the payment amount can convey.

Taxes continue to reveal current government debt. Thus, nonrecipients can recover the previous quarter's aggregate transfer once next-period taxes reveal the resulting debt. Targeting therefore changes both the distribution and timing of information, rather than permanently hiding fiscal policy from nonrecipients.

The eligibility signal requires an extension of the baseline filtering method. Conditional on the perceived law of motion, continuous observations are processed through a Gaussian update. Eligibility then truncates the resulting distribution according to (88). The moments of this truncated distribution are available analytically. I approximate the posterior by a Gaussian distribution with the same mean and covariance, using assumed density filtering following Minka (2001). Unlike in the baseline model, posterior covariance matrices $P_{it|t}$ now differ across households and time, introducing an additional state variable. Intuitively, this arises because the information content of the eligibility signal depends on a household's wage: A low-wage household that believes its wage is far below the median already expects to be eligible and learns little from it. On the other hand, a household that is uncertain whether its wage lies above or below the median learns more from observing the eligibility status. The observation rules out a large range of possible aggregate output values and therefore reduces its posterior uncertainty.

The remaining solution procedure follows the baseline approximate-aggregation approach. For a candidate perceived output law, I solve household decisions using the extended filter, simulate the economy while imposing the within-period equilibrium condition, and update the output-law coefficients. Appendix E provides the filtering equations and evaluates the additional approximation. In these checks, the root mean squared error in output beliefs is 4.4% of households' posterior standard deviation, and the corresponding error in next-quarter eligibility probabilities is 0.09 percentage points. These checks indicate

that the additional filtering approximation introduces small errors in household beliefs and predicted eligibility.

7.3 Results: Full targeted transfer model

The full targeted model preserves the main result of the simplified exercise: Targeting raises the impact consumption response under both information structures, with a larger gain under dispersed information. Table 9 reports the impact responses underlying Figure 15. Under full information, the response rises from approximately 0.38 under universal transfers to 0.68 under the targeted rule. Under dispersed information, it rises from approximately 0.07 to 0.52. Targeting therefore roughly halves the absolute gap between the two information structures. Expressed in relative terms, the effect is even larger. Whereas the full information impact response is more than five times as big as the dispersed information one in the universal regime, it is only about 30% larger in the full targeting model.

Table 9: Impact consumption responses under universal and targeted transfers

Information structure	Universal	Simplified exercise	Full GE model
Full information	0.38	0.51	0.68
Dispersed information	0.07	0.30	0.52

Notes: Approximate aggregate impact consumption responses per dollar of aggregate transfers. The simplified exercise retains baseline household policies and the perceived law of motion. The full model re-solves household decisions and aggregate dynamics under quarterly reassessment of eligibility.

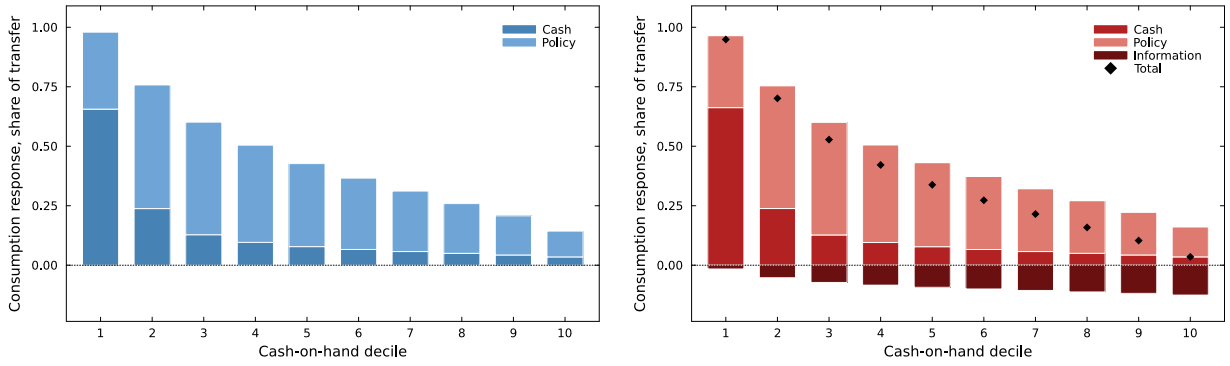
Much of the additional increase relative to the simplified exercise reflects the state contingency of future transfers. With eligibility reassessed every quarter, a fiscal expansion provides support not only to households with low income today, but also to households whose income falls below the cutoff while the expansion persists. Households anticipate receiving payments in low-income states, when marginal utility is high. Relative to payments of the same expected value that do not depend on future income realizations, this support reduces expected future marginal utility and precautionary saving, raising consumption today. The effect also applies to households that are not currently eligible.

To quantify this channel, I replace reassessed eligibility with a schedule of expected payments fixed at impact. Each household receives its expected future transfers conditional on its income at impact. The schedule matches expected payments when it is constructed, but removes their dependence on subsequent household income shocks. Under full information, this counterfactual yields an impact consumption response of approximately

0.51 per dollar, compared with 0.68 under quarterly reassessment. The state contingency of future payments therefore accounts for essentially the entire additional gain over the simplified exercise. For households, state contingency implies additional insurance against negative income shocks during a fiscal expansion.

This insurance effect is also the largest source of the additional increase under dispersed information. Anticipated general-equilibrium feedback plays a more substantial role here, however. Targeting changes the output response more strongly than under full information, and households incorporate the larger response into their expectations about future labor income. This adjustment accounts for more than 10% of the increase in the impact consumption response relative to the simplified exercise.

Figure 17 reports the impact response per dollar received among recipients in each cash-on-hand decile. Cash and policy effects remain positive under both information structures. Under dispersed information, the information component continues to reduce spending, particularly among households with substantial cash on hand. Targeting therefore does not eliminate the information channel, but alleviates its effects on aggregate consumption and output.



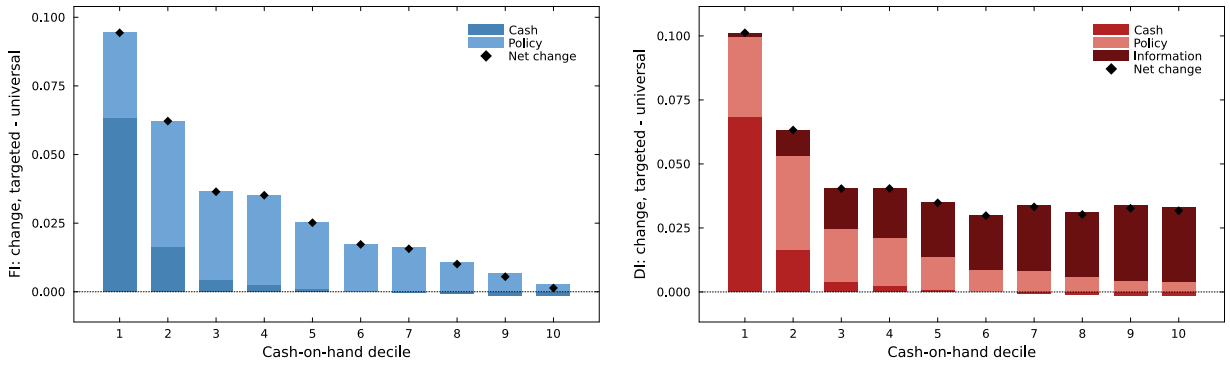
(a) Full information

(b) Dispersed information

Figure 17: Impact consumption responses among recipients in the targeted model

Notes: Average impact consumption responses per dollar received among recipients in each pre-shock cash-on-hand decile, decomposed into cash, policy, and information components.

Figure 18 reports decile contributions to the change relative to universal transfers, including recipients and nonrecipients. Targeting increases consumption contributions in every cash-on-hand decile under both information structures. Under full information, the largest gains occur at the bottom. Under dispersed information, the information component also makes a substantial positive contribution toward the top. The mechanism identified in the simplified exercise therefore survives when households understand the targeted rule and adjust their decisions accordingly.



(a) Full information

(b) Dispersed information

Figure 18: Contributions to the change in the impact consumption response under the targeted rule

Notes: Differences between the targeted and universal-transfer models. Bars report contributions by pre-shock cash-on-hand decile to the change in the aggregate consumption response per dollar of aggregate transfers. Contributions include recipients and nonrecipients.

The targeted model thus speaks to the distinction between resources and information. Targeting directs resources toward households with strong spending responses to cash flows and reduces the effects of the information channel among households whose spending is most sensitive to expectations. Quarterly reassessment additionally provides support against future low-income realizations. While this insurance raises consumption responses under both information structures, the larger gain from targeting under dispersed information remains.

8 Conclusion

Countercyclical transfers affect household spending through both resources and information. When households infer aggregate conditions from their own cash flows, receiving a transfer can signal that aggregate conditions are weak. This paper shows that this (dispersed) information channel can substantially reduce the consumption response to fiscal stimulus. Its importance depends on households' financial resources: Less constrained households respond more strongly to deteriorating expectations and can lower their consumption even as transfers raise their current resources.

Empirical evidence supports this mechanism. A large literature documents that households extrapolate from their own economic circumstances when forming expectations about aggregate conditions. This paper contributes to that literature and links it to evidence on how households react to information about fiscal policy. In a randomized information

experiment, information about a countercyclical fiscal package lowers expected GDP and durable spending intentions. Expectations fall broadly across income groups, while the decline in spending intentions is concentrated among high-income households. These findings support the interaction between information and financial resources central to the model.

In a basic consumption–savings model, inference from cash flows generates hump-shaped aggregate consumption responses to aggregate shocks, while responses to idiosyncratic shocks remain front-loaded. MPCs depend on the information carried by shocks, and consumption responses to transitory shocks can be larger than pure MPCs that hold beliefs fixed. To study this mechanism in a heterogeneous-agent economy with endogenous aggregate output, I develop a global solution method combining Bayesian filtering with approximate aggregation.

The quantitative general-equilibrium model demonstrates the importance of the information channel for fiscal policy. Mapping the information experiment into the model produces a decline in GDP expectations of similar magnitude to that observed in the data and reproduces the direction and heterogeneity of spending responses. In response to a discretionary transfer innovation, the initial aggregate consumption response falls from about 38 cents per dollar under full information to about 7 cents under dispersed information. The information channel accounts for most of this difference and becomes increasingly important toward the top of the cash-on-hand distribution. Households with substantial cash on hand actually lower their consumption upon receiving a transfer, as the adverse information outweighs the positive cash and anticipated policy effects. This heterogeneity reflects stronger spending responses to deteriorating expectations among less constrained households, consistent with the experimental evidence. The results also help account for suggestive evidence that large countercyclical payment programs induce smaller consumption responses than similarly sized procyclical or acyclical payments.

Lastly, I find that targeting transfers toward low-income households becomes much more effective under dispersed information. While targeting already increases the output response under full information, the information channel amplifies this gain. Targeting directs resources toward households whose consumption responds strongly to current cash flows and less strongly to the adverse information conveyed by transfers. The effectiveness of fiscal transfers thus depends on both the resources they provide and the aggregate conditions households infer from receiving them.

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A Responses to fiscal transfers: empirical benchmarks

This appendix collects micro- and macro-level consumption benchmarks to complement the empirical evidence and model results.

A.1 Household MPCs and the policy context

Figure 19 compares selected quarterly MPC estimates from U.S. studies. Responses to regular tax refunds and Alaska dividends exceed the estimates for three countercyclical episodes. Due to differences in settings and underlying data, this comparison is suggestive rather than causal.

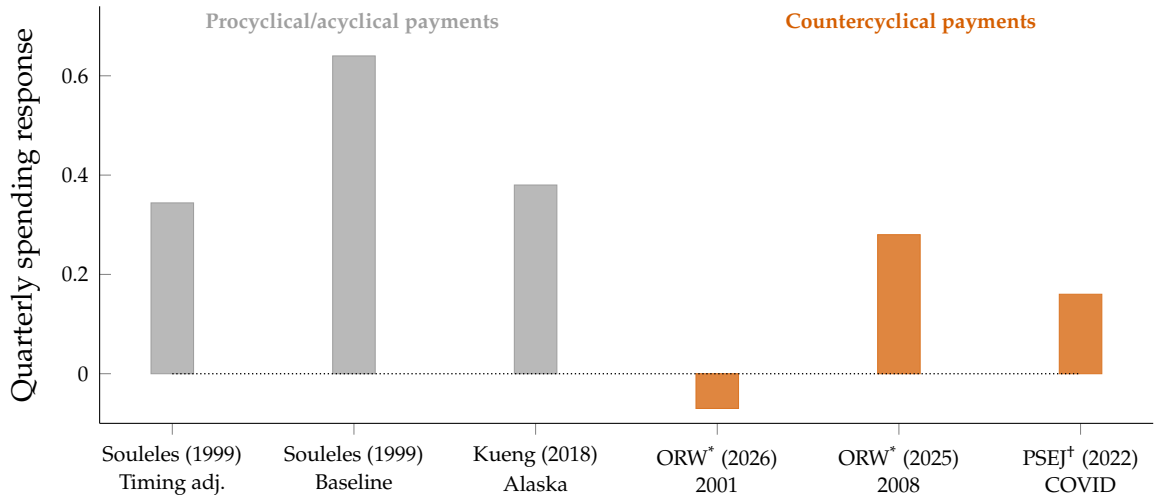


Figure 19: Selected quarterly household MPC estimates by policy context

Notes: Bars show selected point estimates. All estimates are total spending responses including durables. Souleles values are alternative specifications from the same study. The Kueng value combines the quarterly responses of nondurable and service spending (0.25) and classified durable purchases (0.13). The negative 2001 point estimate does not establish a significantly negative response. *ORW denotes Orchard, Ramey, and Wieland; †PSEJ denotes Parker, Schild, Erhard, and Johnson. The PSEJ estimate averages over the coefficients for the three EIP rounds they consider. Cyclicality labels refer to the policy context and design, rather than an estimated correlation between payments and output.

Related evidence comes from Goldfayn-Frank et al. (2022), who estimate an overall low MPC of 5.4% across the three countercyclical *Kinderbonus* payments in Germany during COVID, although Bachmann et al. (2022) find much larger responses for the first two installments. Conversely, Agarwal and Qian (2014) find a large spending response of about 0.80 per dollar from Singapore's procyclical Growth Dividend. Both studies are excluded because they concern non-U.S. settings and non-quarterly horizons: the Kinderbonus estimates use spending within the payment months, while Singapore's ten-month estimate starts at the announcement, even including spending before disbursement.

The earlier rebates' lower MPCs are not explained by payment size. Fagereng et al. (2021)

document that MPCs decline with lottery prize size. Yet Souleles (1999) reports a mean household refund of $874 in 1982–1984 dollars, and Kueng (2018) reports average Alaska dividends of $1,650 per person, or $4,600 per household, in 2014 dollars. The 2001 rebates were generally $300–$600 per household, and the 2008 payments averaged about $1,000 per household. In essence, if anything, differences in payment size would bias the procyclical or acyclical MPCs downward.

The pattern is noteworthy given that there is evidence of state dependence in MPCs. Sokolova (2023) finds that MPC estimates increase with unemployment rates, while Gross et al. (2020) find higher MPCs out of additional credit availability during recessions and in areas with higher unemployment. The information mechanism pushes the other way: countercyclical payments confer negative news that can offset stronger spending responses associated with weak economic conditions.

A.2 Aggregate income and consumption around stimulus payments

Figure 20 plots real disposable income and consumption around the 2001 and 2008 U.S. tax rebates. Temporary income spikes do not come with consumption increases. This is suggestive, rather than causal evidence. Nevertheless, these patterns motivate examining why stimulus payments can generate modest aggregate consumption responses, as done by Taylor (2009, 2022), Orchard, Ramey, and Wieland (2025, 2026), and Ramey (2026).

The divergence is even more pronounced during the COVID-19 pandemic (Figure 21). Restrictions, health risks, and other contemporaneous policies directly affected spending, however. The figure illustrates the divergence between aggregate income and consumption. It does not establish that the payments themselves had a small or negative effect on consumption.

B Additional Empirical Results

This appendix provides additional information on the construction of the SCE and BOP-HH samples and reports supplementary results supporting the empirical analysis in the main text. The experience-effects appendix documents the full specifications behind the SCE core-panel results and the household-income evidence from the BOP-HH. The experimental appendix reports validity checks, raw purchase-probability responses, income heterogeneity, the persistence of the fiscal-package treatment effect, and savings-related outcomes. Appendix D describes the model validation, including the conversion of durable-purchase intentions into consumption revisions and the household-income comparison.

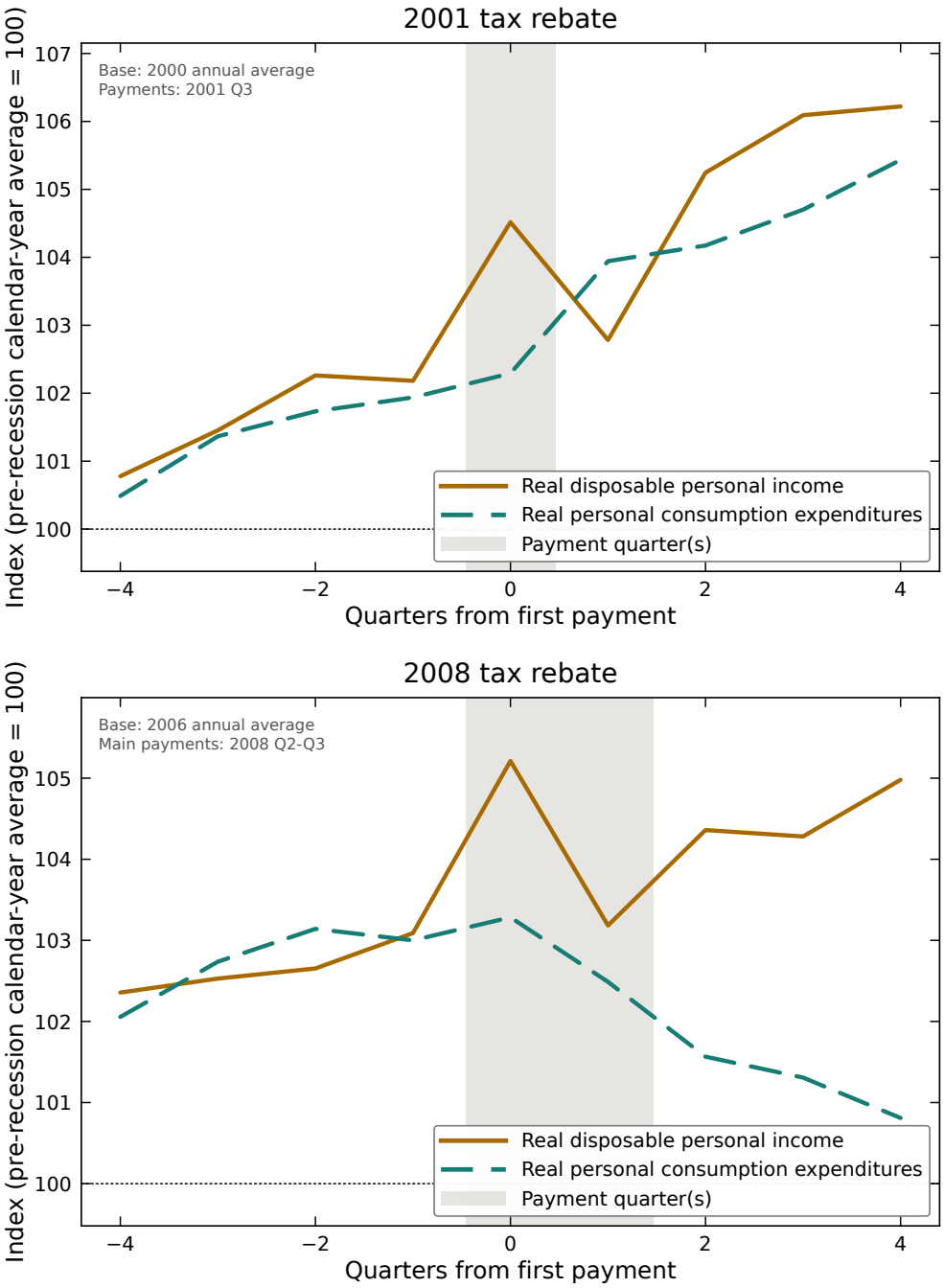


Figure 20: Real disposable income and consumption around the 2001 and 2008 tax rebates

Notes: Quarterly real disposable personal income (DPIC96, BEA account code A067RX) and real personal consumption expenditures (PCECC96, DPCERX), from the BEA National Income and Product Accounts, retrieved through FRED (U.S. Bureau of Economic Analysis, n.d.-a; U.S. Bureau of Economic Analysis, n.d.-b). The underlying series are seasonally adjusted annual rates in billions of chained 2017 dollars. Each series is indexed to its own calendar-year average: 2000 for the 2001 episode and 2006 for the 2008 episode. Quarter zero is 2001 Q3 and 2008 Q2, respectively. Shading marks payment quarters. Vertical scales differ across panels.

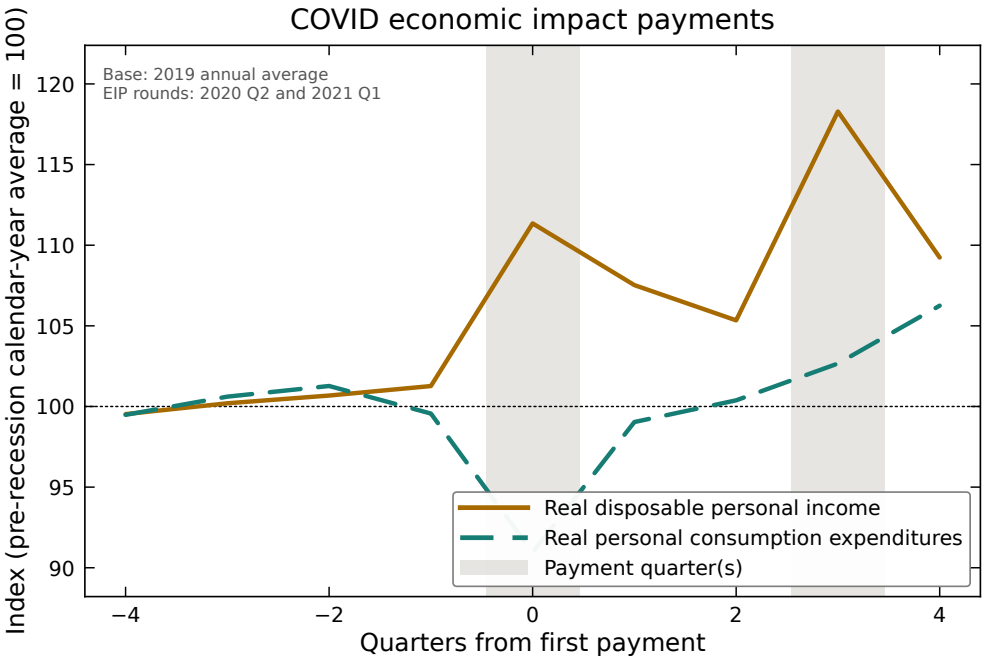


Figure 21: Real disposable income and consumption around COVID economic impact payments

Notes: Data sources and units are as in Figure 20 (U.S. Bureau of Economic Analysis, n.d.-a; U.S. Bureau of Economic Analysis, n.d.-b). Each series is indexed to its own 2019 calendar-year average. Quarter zero is 2020 Q2. Shading marks the payment quarters shown in the figure: 2020 Q2 and 2021 Q1.

B.1 Data and measurement

B.1.1 SCE core panel

I use the SCE core panel between June 2013 and September 2025 to study the relationship between households' financial circumstances and their expectations about aggregate unemployment. The household financial-situation variable is question Q1, which asks whether the household is financially better or worse off than twelve months earlier on a five-point scale from 1, much worse, to 5, much better. Aggregate unemployment expectations are measured by Q4new, the reported percent chance that the US unemployment rate will be higher twelve months later.

The analysis sample retains observations with nonmissing Q1, Q4new, and positive survey weights. The main fixed-effects regressions use SCE survey weights and cluster standard errors by respondent.

For the rationality exercise, I define the forecast as

$$F_{it} = Q4new_{it}/100. \tag{89}$$

The realization is

$$R_{it} = \mathbb{1}\{u_{t+12} > u_t\}, \tag{90}$$

where u_t is the US unemployment rate in the interview month. The forecast error is

$$FE_{it} = R_{it} - F_{it}. \tag{91}$$

The rationality sample excludes interviews for which the unemployment rate twelve months later is unavailable. The primary covariance estimator clusters by both survey month and respondent. This accounts for the fact that the realization is common to all respondents interviewed in the same month, while forecast and reporting errors may also be correlated within respondents.

B.1.2 BOP-HH experience panel

The BOP-HH experience sample uses the Scientific Use File (SUF), version 2025Q4 (Deutsche Bundesbank, 2025), and covers waves 4 through 69, corresponding to April 2020 through September 2025. Aggregate beliefs are measured using qualitative questions about whether German unemployment and economic growth will decrease, remain approximately unchanged, or increase over the following twelve months.

Household income is reported in brackets and converted to the midpoint supplied with the scientific-use file. Nominal household-income growth is the log change in this midpoint between consecutive interviews. Real income growth subtracts German HICP inflation over the corresponding interval. Both nominal and real growth are winsorized at their respective first and ninety-ninth percentiles.

The BOP-HH variable refers to total household income and remains defined following a job change or job loss, making it well suited to measuring adverse changes in household resources. One significant issue is that the survey is mostly only available during a time with high (expected) inflation. This together with the measurement of income through brackets or midpoints means many wage increases likely go unmeasured, while movements crossing brackets that are recorded might simply reflect inflation adjustments, possibly even decreases in real wages.

The SCE and BOP-HH aggregate-unemployment variables measure similar but not identical objects. The SCE asks for the probability that unemployment will be higher, while the BOP-HH asks for the direction of change on a qualitative scale. In the shared experience-effect specifications, belief outcomes are scaled by their full-sample standard deviations.

B.1.3 BOP-HH randomized experiment

The randomized information experiment was conducted in wave 4 of the BOP-HH in April 2020 (Deutsche Bundesbank, 2020). Respondents were assigned to one of six treatment arms:

1. placebo information about European human rights;
2. recession news;
3. uncertainty news;
4. information about the ECB Pandemic Emergency Purchase Programme;
5. information about the German federal fiscal package;
6. government reassurance.

The empirical analysis in the paper compares respondents assigned to the federal fiscal-package treatment, arm 5, with respondents assigned to the placebo treatment, arm 1. Arms 2, 3, 4, and 6 are excluded from the main treatment regressions. The treatment indicator is therefore

$$fp_i = \mathbf{1}\{\text{respondent } i \text{ is assigned to arm } 5\}, \tag{92}$$

within the sample consisting of arms 1 and 5.

Post-treatment GDP expectations are constructed from the eight-bin probability distribution in question 419. The mean forecast uses the bin midpoints and is expressed in percentage points of expected GDP growth. Spending outcomes are stated propensities from 0 to 100 to purchase a house, purchase a car, purchase another large durable good, or build up additional financial reserves over the following twelve months. Appendix B.7 discusses the interpretation of the reserves question.

All experimental regressions use BOP-HH survey weights and heteroskedasticity-robust standard errors. The standard control set includes gender, age, household size, employment status, and pre-treatment qualitative growth expectations. Treatment also affects several other indicators for expectations about aggregate and individual conditions going forward. Table 10 collects these expectations. Broadly, these support a pessimistic revision of expectations due to the fiscal policy information, although many indicators are noisy. For some, this is due to being reported one wave later, with attrition in the panel cutting the number of respondents roughly by half.

The four May outcomes in Table 10 additionally control for the April rating of the economy as a serious problem (Q410e). The income-expectation regressions use the post-treatment forecast as the dependent variable and additionally control for the corresponding pre-treatment forecast. Expected monthly euro changes are constructed from the twelve probability bins, divided by the midpoint of current monthly household net income, and multiplied by 100. The main income sample is untrimmed. Open income-change tails are represented by $\pm$2,250 euros and the two bins adjacent to zero by $\pm$125 euros; there is no separate zero-change bin. Respondents are instructed to assume unchanged taxes and social contributions. For the GDP density, the open tails are represented by $\pm$17.5%.

B.2 Experience effects and forecast efficiency

B.2.1 Financial conditions and unemployment expectations

The following table reports the full set of specifications relating respondents' household financial situation to their aggregate unemployment expectations. It includes the pooled specification, month fixed effects, respondent fixed effects, and both sets of fixed effects, with weighted and unweighted estimates.

Table 11: Own financial situation and aggregate unemployment expectations (SCE)

	Unweighted	Survey-weighted
No fixed effects	-4.128***	-3.943***
	(0.121)	(0.160)
Month fixed effects	-3.893***	-3.759***
	(0.122)	(0.161)
Respondent fixed effects	-1.858***	-1.884***
	(0.097)	(0.132)
Respondent and month fixed effects	-1.577***	-1.600***
	(0.095)	(0.128)
Observations	185,086	

The relationship between household financial conditions and aggregate unemployment expectations is negative across specifications. The month-fixed-effects estimates compare respondents interviewed under the same aggregate conditions, while the respondent-fixed-effects estimates use changes within respondents during their SCE panel spells.

B.2.2 Forecasts, realizations, and forecast errors

The following table reports the full rationality exercise. In addition to the preferred two-way-clustered standard errors, it includes specifications clustering only by survey month and only by respondent.

The estimated financial-situation slopes for forecasts and realizations are quantitatively similar. The corresponding forecast-error slope is close to zero and is not statistically distinguishable from zero under two-way clustering. Respondent-only clustering understates uncertainty in the realization equation because the realized aggregate outcome varies only across survey months.

B.2.3 BOP-HH household-income changes

Figure 22 reports the asymmetric relationship between real household-income changes and revisions in aggregate expectations. Positive and negative income changes enter the same regression with a shared intercept, wave fixed effects, and a control for the number of waves since the previous interview.

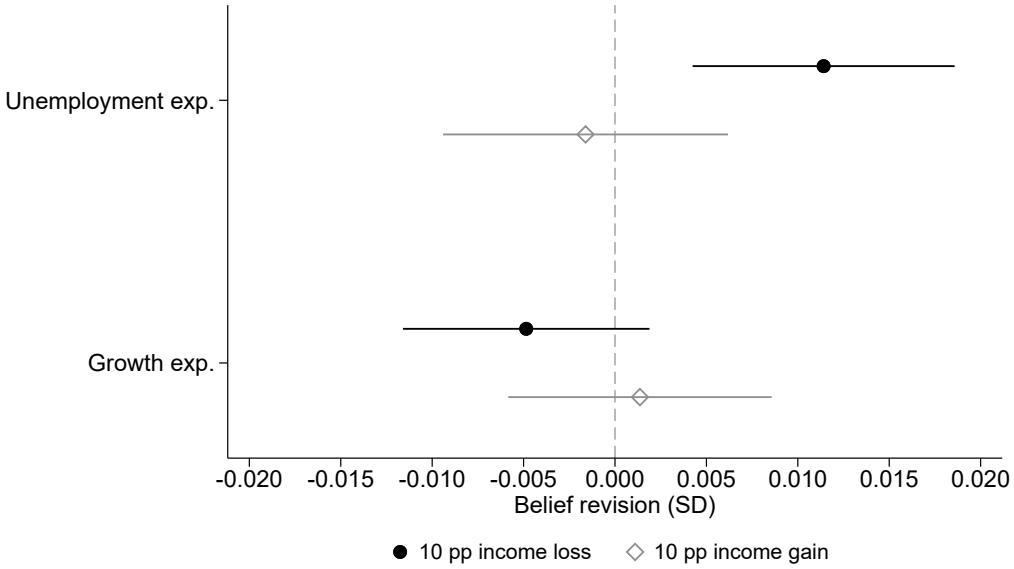


Figure 22: Real household-income changes and aggregate expectations in the BOP-HH

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020–September 2025, own calculations.

The point estimates imply larger responses to real-income losses than to gains. Tests of equality between the gain and loss slopes yield $p = 0.063$ for unemployment expectations and $p = 0.492$ for growth expectations. The growth response to income losses is itself imprecisely estimated ($p = 0.158$). The evidence for asymmetry is therefore suggestive for unemployment and weak for growth.

B.3 Experimental validity

B.3.1 Covariate balance

Table 13 reports balance in pre-treatment covariates between respondents assigned to the fiscal-package treatment and respondents assigned to the placebo treatment.

The unweighted balance tests evaluate whether random assignment generated comparable groups in the arm-5-versus-arm-1 sample. The weighted tests provide a corresponding diagnostic for the weighted estimation sample. Survey weights were not used in treatment

assignment, so differences between weighted and unweighted balance tests do not imply a failure of randomization.

B.3.2 GDP expectations across experimental arms

Figure 23 reports mean post-treatment GDP expectations in all six experimental arms.

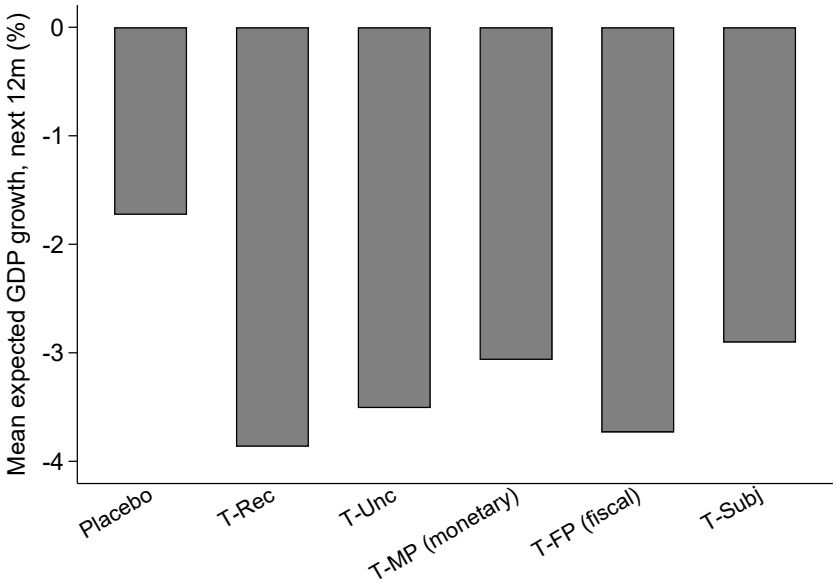


Figure 23: Post-treatment GDP expectations by experimental arm

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

Although the main regressions compare only the fiscal-package and placebo arms, the six-arm figure provides a transparent overview of the experimental variation. It also validates the construction of the post-treatment GDP-growth forecast from the probability bins.

B.4 Spending heterogeneity

B.4.1 Income-bracket profiles

Figure 24 reports the effects of the fiscal-package treatment on expected GDP growth and intended durable purchases separately across the five household-income groups. Each coefficient compares arm 5 with placebo arm 1.

Table 10: Treatment effect on other expectations

Outcome	Item / wave	Units	ATE	SE	p-value	N
Economy as a serious problem	Q515e May 2020	score points	+0.447	0.189	0.0187	329
Concern about the German economy	Q518a May 2020	score points	+0.238	0.236	0.3135	331
Concern about the EU economy	Q518b May 2020	score points	+0.288	0.277	0.2988	331
Concern about own future finances	Q518d May 2020	score points	+0.622	0.391	0.1121	331
Expected stock-market return	Q422 April 2020	pp	-1.051	1.663	0.5275	593
Expected local house-price growth	Q420 April 2020	pp	+1.579	1.074	0.1419	636

Notes: Problem and concern outcomes are measured on 1–10 scales; their effects are changes in mean ratings. Reported p-values are nominal and the outcomes are exploratory. The economy-as-a-problem estimate has $p = 0.019$ before adjustment and $p = 0.112$ after a Holm correction across the six outcomes in this table.

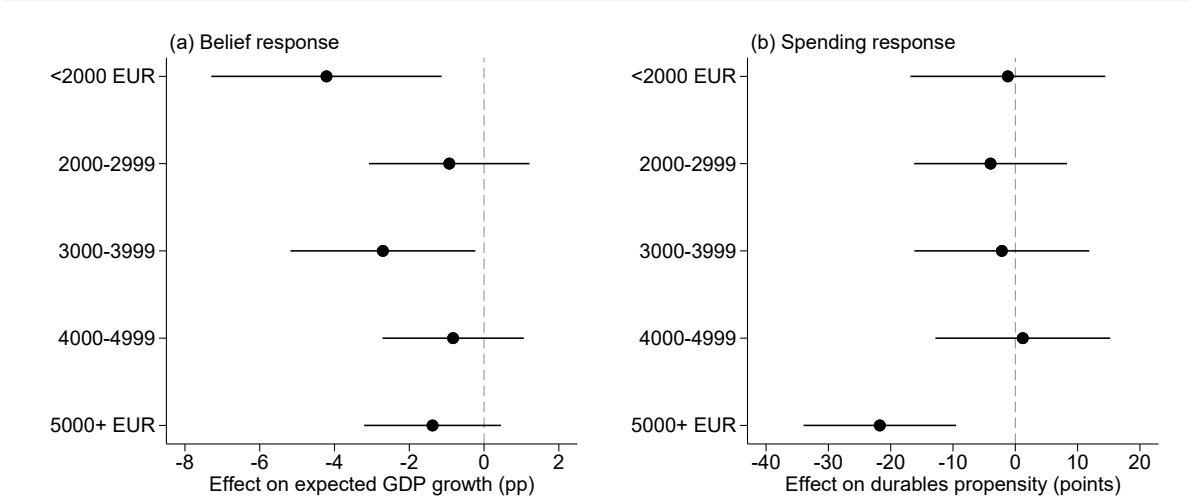


Figure 24: Fiscal-package information, GDP expectations, and intended durable spending by household income

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

Table 12: Forecast efficiency: stated probability versus realized outcome

	Coefficient	Two-way SE	Month SE	Respondent SE
Forecast	-0.0391***	(0.0021)	(0.0016)	(0.0016)
Realization	-0.0362***	(0.0105)	(0.0100)	(0.0034)
Forecast error	0.0029	(0.0108)	(0.0102)	(0.0038)
Observations		180,863		
Survey months		144		
Respondents		24,049		

B.4.2 Purchase-probability responses

Table 14 reports the raw durable-purchase-probability responses underlying the spending comparison. These outcomes are stated intentions rather than expenditure amounts. Section 6.1 reports consumption revisions constructed using the category-specific purchase-probability responses, control-group probabilities, and expenditure weights described in Appendix D.6.

B.5 Persistence of the fiscal-package belief effect

Figure 25 reports the effect of the fiscal-package information treatment on qualitative growth expectations in the months following the experiment.

The figure shows whether the initial effect of the fiscal-package treatment remains visible in the subsequent BOP-HH waves. Precision declines over the follow-up period because of rapid panel attrition and because the analysis is restricted to the fiscal-package and placebo arms.

The follow-up outcome is the qualitative five-category growth-expectations question from the regular BOP-HH module. It is not the original post-treatment GDP-density forecast, which was not repeatedly elicited for the surviving experimental households. The persistence exercise therefore examines whether the treatment continues to affect a closely related measure of aggregate growth expectations rather than the exact original outcome. The large standard errors are partly due to attrition: Between treatment and control group, there are only 331 respondents left in wave 5, 160 in wave 6 and 111 in wave 7.

B.6 Prior belief relevance for the effect

Figure 26 shows that the reduction in intended spending is concentrated among respondents who were relatively optimistic before receiving the fiscal-package information. For these respondents, the announcement represents more adverse news relative to their prior beliefs. The saving-intention estimates are imprecise and do not provide clear evidence of increased precautionary saving. Appendix B.7 discusses the measurement and

Table 13: Randomization balance: placebo versus selected treatment arms

Covariate	Placebo	Policy	Difference	p-value	N
Panel A: Full analysis sample					
Age	53.358	54.088	0.730	0.557	681
Household size	2.248	2.232	-0.015	0.854	679
Gender	1.510	1.444	-0.066*	0.084	681
Employed	0.571	0.566	-0.004	0.912	679
Homeowner	0.683	0.641	-0.042	0.246	681
Income bracket	2.868	2.811	-0.057	0.584	647
Prior growth outlook	1.707	1.605	-0.102	0.173	680
Credit constrained	0.091	0.135	0.044*	0.070	680
Joint F-test (Unweighted)	–	–	1.398	0.194	645
Joint F-test (Weighted)	–	–	1.902	0.057	645
Panel B: Top-income group					
Age	48.800	52.873	4.073	0.202	100
Household size	3.022	2.745	-0.277	0.173	100
Gender	1.356	1.309	-0.046	0.628	100
Employed	0.689	0.673	-0.016	0.865	100
Homeowner	0.822	0.764	-0.059	0.475	100
Income bracket	5.000	5.000	0.000	–	100
Prior growth outlook	1.756	1.582	-0.174	0.397	100
Credit constrained	0.067	0.109	0.042	0.456	100
Joint F-test (Unweighted)	–	–	0.869	0.534	100
Joint F-test (Weighted)	–	–	1.062	0.395	100

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

interpretation of this outcome.

B.7 Savings intentions and financial constraints

The April 2020 survey asks respondents for the probability that they will "build up additional financial reserves" over the following twelve months. The estimated treatment effect is negative, at -3.58 percentage points, but statistically insignificant. The income gradients are also imprecisely estimated. This question does not distinguish longer-term asset accumulation from the accumulation of liquid funds available to cover near-term expenses.

The May 2020 survey instead asks whether respondents have or anticipate difficulty covering current expenditure in the near future because of the coronavirus pandemic. Table 15 reports effects on the probability of financial constraints over one- and three-month horizons. The point estimates are negative, indicating less anticipated difficulty financing expenditure. The one-month estimates are more precise; the three-month results are more sensitive to the control specification. The smaller samples reflect attrition between the April experiment and the May follow-up.

These outcomes concern different aspects of household finances. Respondents may inter-

Table 14: Raw durable-purchase-probability responses

	Effect	Standard error
Average treatment effect	−6.15	3.09
Slope across income brackets	−2.27	2.22
Slope across prior categories	−4.53	2.97

Notes: Entries are treatment effects on stated durable-purchase propensity, in percentage points. Slope rows report the change in the treatment effect per one-category increase in household income or prior optimism about growth. Estimates compare the fiscal-package treatment with the placebo and use survey weights and the experimental control set described in Appendix B.1.3.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

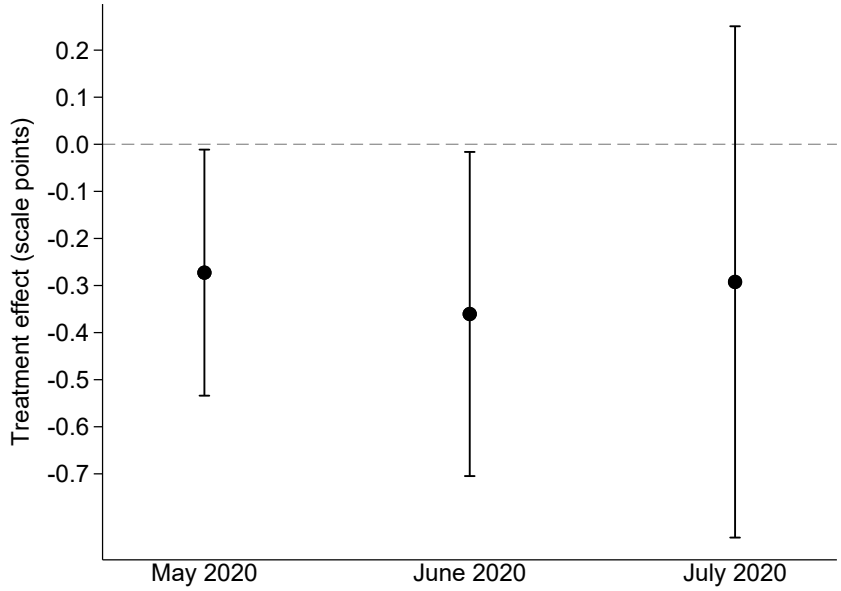


Figure 25: Persistence of the fiscal-package treatment effect on growth expectations

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April–July 2020, own calculations.

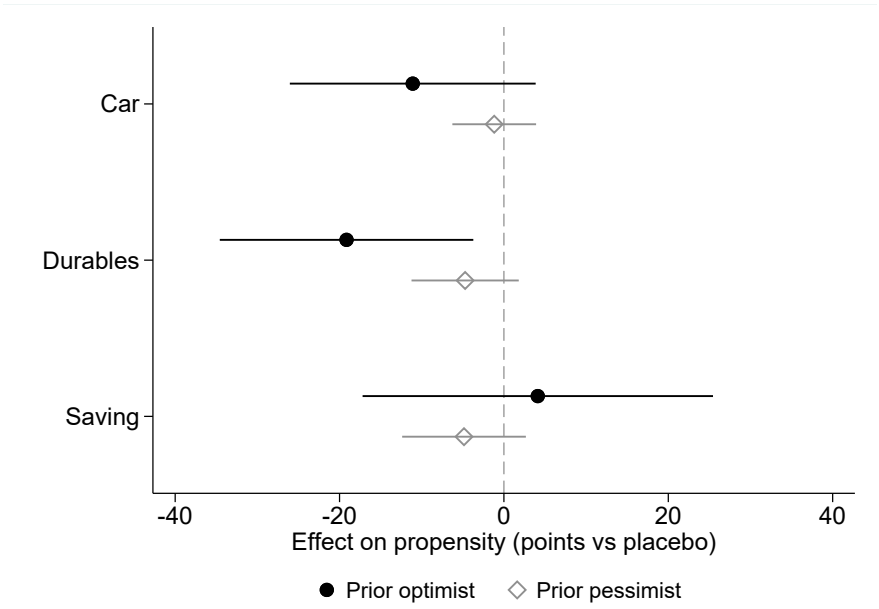


Figure 26: Spending and saving responses by prior growth expectations

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

pret building reserves as longer-term saving, while near-term financing capacity depends on income and liquid funds already available. A lower reported probability of building reserves can therefore coexist with an improved ability to cover expenses. The follow-up responses do not establish that households increased realized saving. I consequently report both outcomes but do not use either as a quantitative counterpart to saving in the model.

B.8 Construction of untargeted belief moments

This appendix describes the construction of the belief moments reported in Table 7. The empirical moments are calculated using wave 68 of the Bundesbank Online Panel – Households (BOP-HH), conducted in August 2025. The corresponding model moments are calculated without changing the baseline calibration.

Survey measurement. Question P6809 asks respondents to assess the percentage change in German real GDP in Q2 2025 relative to Q1 2025. The question is asked before respondents are shown the published GDP figure in the subsequent question. The response therefore provides a measure of respondents' assessments of past aggregate growth, rather than a forecast of future growth.

Responses are recorded in seven categories, distinguishing declines and increases of different magnitudes, as well as approximately unchanged output. To recover a measure of cross-sectional dispersion, I fit an interval-normal model to these categorical responses. The three categories closest to zero are combined into an interval from -0.25% to 0.25%.

Table 15: Savings intentions and anticipated financial constraints

Estimand / specification	Effect	SE	p-value	N
Build additional financial reserves: April 2020				
Average treatment effect	−3.58	3.61	0.322	637
Slope per EUR 1,000 of income	1.20	1.86	0.520	611
Slope per pooled income bracket	2.22	2.74	0.419	611
Probability of financial constraints: May 2020				
One month: core controls	−5.70	3.34	0.0885	329
One month: add April liquidity and income	−8.00	3.23	0.0138	318
Three months: core controls	−1.71	4.37	0.6962	329
Three months: add April liquidity and income	−7.17	3.72	0.0551	318

Notes: Effects are in percentage points; income slopes are per EUR 1,000 or per one-category increase, as indicated. Estimates compare the fiscal-package treatment with the placebo and use survey weights. The extended financial-constraints specifications additionally control for April liquidity and income. SE denotes standard error.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April–May 2020, own calculations.

The remaining categories distinguish changes between 0.25% and 0.50%, with open-ended tails beyond 0.50%. Category boundaries are transformed from percentage changes g into log percentage points using $100 \log(1 + g/100)$.

To measure the relationship between household income and aggregate beliefs, I link respondents to their household-income reports in wave 62, conducted in February 2025. Household income is reported in brackets, which I represent by their assigned midpoints. The resulting six-month change in log household net income is denoted by x_i. I estimate the conditional mean of the latent GDP-growth assessment as

$$\mu_i = \alpha + \beta x_i + \gamma \log I_{i,62} + Z_i'\delta, \tag{93}$$

where $I_{i,62}$ denotes the initial household-income midpoint. The controls Z_i include age, age squared, gender, household size, children, and education. The latent GDP-growth assessment is assumed normally distributed around μ_i, with residual variance σ_B^2.

I estimate the model by maximizing the survey-weighted interval-normal likelihood, using the reported GDP-growth categories as interval observations. Confidence intervals are based on heteroskedasticity-robust standard errors. The estimated slope is $\hat{\beta} = 0.181$, with a 95% confidence interval of $[0.067, 0.294]$.

The corresponding estimate of total cross-sectional belief dispersion is

$$\widehat{\mathrm{SD}}(B_i) = \sqrt{\hat{\sigma}_B^2 + \mathrm{Var}_w(\hat{\mu}_i)}, \tag{94}$$

where Var_w denotes the survey-weighted cross-sectional variance. This yields a standard deviation of 0.274 log percentage points in the six-month matched sample.

Importantly, this estimate is based on a parametric reconstruction of latent assessments from categorical responses. It may contain reporting variation in addition to genuine disagreement about aggregate conditions. These components cannot be separately identified from the observed responses.

Model counterparts. The survey question concerns past GDP growth, whereas the model's information structure primarily generates heterogeneity in beliefs about current aggregate output. I therefore construct two distinct moments.

Let $\hat{y}_{i,t-k|t}$ denote household i's posterior estimate of log aggregate output in quarter $t-k$, conditional on all information available through quarter t. The model counterparts of the survey backcast and current-output belief are

$$b_{it} = 100\left(\hat{y}_{i,t-1|t} - \hat{y}_{i,t-2|t}\right), \tag{95}$$

$$m_{it} = 100\hat{y}_{i,t|t}. \tag{96}$$

The interview quarter corresponds to t, so the GDP-growth backcast concerns output in quarters $t-1$ and $t-2$. The historical posterior estimates are obtained using the model's linear-Gaussian filtering system, incorporating information received through quarter t. No additional public GDP signal is imposed.

To approximate the survey's measure of household income, which asks respondents to report average monthly income over the preceding twelve months if income varies, I calculate a four-quarter moving average of model household net income:

$$J_{it} = \frac{1}{4}\sum_{j=0}^{3} I_{i,t-j}. \tag{97}$$

I then regress b_{it} on the six-month change $\log J_{it} - \log J_{i,t-2}$, controlling for $\log J_{i,t-2}$. Unlike the survey measure, model income is observed continuously rather than through income brackets. The resulting slope is 0.120. The corresponding twelve-month specification gives a slope of 0.033.

The model moments are computed across 200,000 simulated households, conditional on a common aggregate state at its reference level. Under dispersed information, the cross-sectional standard deviations of current-output beliefs and GDP-growth backcasts are 0.195 and 0.032 log percentage points, respectively. Under full information, both moments and the income-belief slope are zero.

Measurement sensitivity. Table 16 reports alternative survey measurements. The twelve-month specification uses household income reported in wave 56 instead of wave 62. The shorter- and longer-horizon samples are not identical.

I also report estimates obtained by assigning representative values to the categorical GDP-growth responses and estimating the regression by weighted least squares. These estimates vary the representative values assigned to the open-ended response categories. They illustrate sensitivity to the measurement of categorical beliefs and should not be interpreted as confidence bounds on underlying disagreement.

Table 16: Sensitivity of untargeted survey belief moments

Specification	N	SD	Slope	95% CI
Six-month income change, baseline	2,306	0.274	0.181	[0.067, 0.294]
Twelve-month income change	1,332	0.277	0.039	[−0.075, 0.154]
Six months, midpoint coding (±0.50)	2,306	0.245	0.100	[−0.001, 0.202]
Six months, midpoint coding (±0.75)	2,306	0.302	0.135	[0.012, 0.257]
Six months, midpoint coding (±1.00)	2,306	0.370	0.169	[0.023, 0.314]

Notes: Standard deviations and slopes are in log percentage points. Baseline specifications use the survey-weighted interval-normal estimator and control for starting household income, age, age squared, gender, household size, children, and education. Confidence intervals are heteroskedasticity-robust. Midpoint specifications use weighted least squares with alternative representative values, in percent, for the open-ended GDP-growth categories. These are measurement assumptions, not confidence bounds.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, BOP-HH waves 56, 62, and 68, own calculations.

The positive income-belief relationship is broadly robust to these alternative measurements, although its magnitude varies. By contrast, the level of measured disagreement depends substantially on assumptions used to reconstruct beliefs from categorical responses. The dispersion moment should therefore be interpreted as a descriptive benchmark, rather than a precise target for the model's information structure.

C Machine-learning signal-extraction counterfactuals

This appendix describes the alternative forecasting rules used in the ex-post signal-extraction exercise. All methods forecast current aggregate log output y_t from household-observable variables. Define the current observation vector as

$$o_{it} = (w_{it}, \widetilde{T}_t, D_t)'. \tag{98}$$

Here, D_t is observed exactly through the tax rule. All standardization parameters and forecasting coefficients are estimated using the training sample only. Hyperparameters are selected using expanding-window cross-validation over time blocks.

C.1 Ridge regression

The Ridge specification uses the most recent L observations of all three observable series. Define

$$\mathbf{s}_{it} = \left(o_{it}', o_{i,t-1}', \ldots, o_{i,t-L+1}'\right)' \in \mathbb{R}^{3L}, \qquad L = 12. \tag{99}$$

Equivalently, $\mathbf{s}_{it}$ contains the current and lagged values of w_{it}, $\widetilde{T}_t$, and D_t. The inputs are standardized using training-sample means m_s and standard deviations d_s:

$$\widetilde{\mathbf{s}}_{it} = (\mathbf{s}_{it} - m_s) \oslash d_s. \tag{100}$$

The Ridge forecast is

$$\widehat{y}_{it}^{\text{ridge}} = \alpha + \beta' \widetilde{\mathbf{s}}_{it}. \tag{101}$$

The parameters solve

$$(\widehat{\alpha}, \widehat{\beta}) = \arg\min_{\alpha,\beta} \sum_{(i,t)\in\mathcal{T}} \left(y_t - \alpha - \beta' \widetilde{\mathbf{s}}_{it}\right)^2 + n_{\mathcal{T}} \lambda \|\beta\|^2, \tag{102}$$

where $\mathcal{T}$ denotes the training sample and $n_{\mathcal{T}}$ is its number of observations. The penalty parameter λ is selected by expanding-window cross-validation.

The current observation vector is an input rather than a predetermined household state. The recursive state required to carry the forecasting rule forward therefore consists of the preceding $L-1$ observations of each series, for a total of

$$3(L-1) = 33 \tag{103}$$

state variables per household.

C.2 Gated recurrent unit

The GRU replaces the finite observation window with a nonlinear recursive state $h_{it} \in \mathbb{R}^H$. Let $\widetilde{o}_{it}$ denote the standardized current observation vector and initialize $h_{i0} = 0$. The recursion is

$$r_{it} = \Lambda\left(W_r \widetilde{o}_{it} + U_r h_{i,t-1} + b_r\right), \tag{104}$$

$$g_{it} = \Lambda\left(W_g \widetilde{o}_{it} + U_g h_{i,t-1} + b_g\right), \tag{105}$$

$$\widetilde{h}_{it} = \tanh\left(W_h \widetilde{o}_{it} + r_{it} \odot (U_h h_{i,t-1}) + b_h\right), \tag{106}$$

$$h_{it} = (1 - g_{it}) \odot \widetilde{h}_{it} + g_{it} \odot h_{i,t-1}. \tag{107}$$

Here, $\Lambda(v) = 1/(1+\exp(-v))$ is applied element by element and $\odot$ denotes elementwise multiplication. The output forecast is

$$\widehat{y}_{it}^{\text{GRU}} = \mu_y + \sigma_y\left(a' h_{it} + c\right), \tag{108}$$

where μ_y and σ_y are computed from the training sample.

The parameters minimize training-sample mean squared error using truncated backpropagation through time. The hidden-state dimension is selected by validation from

$$H \in \{8, 16\}. \tag{109}$$

Validation selects $H = 16$. The forecasting rule therefore requires sixteen recursively updated state variables per household.

C.3 k-nearest neighbors

The kNN estimator uses the same standardized L-period vector $\widetilde{\mathbf{s}}_{it}$ as the Ridge specification. Let the reference sample be

$$\mathcal{R} = \left\{ \left(\widetilde{\mathbf{s}}_j, y_j \right) \right\}_{j=1}^{N}, \qquad N = 25{,}000. \tag{110}$$

For a household-period observation $\widetilde{\mathbf{s}}_{it}$, define the distance to reference observation j as

$$d_j = \left\| \widetilde{\mathbf{s}}_{it} - \widetilde{\mathbf{s}}_j \right\|_2. \tag{111}$$

Let $\mathcal{N}_k(\widetilde{\mathbf{s}}_{it})$ contain the indices of the k smallest distances. The inverse-distance weights and output forecast are

$$\omega_j = \frac{1}{d_j + \epsilon_d}, \qquad \widehat{y}_{it}^{k\mathrm{NN}} = \frac{\sum_{j \in \mathcal{N}_k(\widetilde{\mathbf{s}}_{it})} \omega_j y_j}{\sum_{j \in \mathcal{N}_k(\widetilde{\mathbf{s}}_{it})} \omega_j}. \tag{112}$$

The constant $\epsilon_d > 0$ prevents division by zero. The number of neighbors is selected by expanding-window cross-validation, which yields $k = 160$.

As with Ridge, the household must retain $3(L-1) = 33$ lagged observations. In addition, implementation of the forecasting rule requires storing and searching the reference sample $\mathcal{R}$.

D Design of the fiscal-announcement validation

This appendix describes how households interpret the cumulative fiscal announcement used in Section 6.1. For this validation exercise, Y^{ref} is Q1 2020 output. The announcement covers Q2 2020 through Q1 2021 and reveals the normalized four-quarter total

$$s = \sum_{j=0}^{3} \widetilde{T}_{t+j} = 0.1696. \tag{113}$$

It does not reveal separate quarterly transfers.

D.1 The announcement as a signal

For this exercise, it is convenient to use the augmented household state

$$q_{it}^a = (\widetilde{y}_t, e_{it}, \nu_t, D_t)'. \tag{114}$$

Let its perceived transition be

$$q_{i,t+1}^a = a_q + A^a q_{it}^a + G^a \varepsilon_{i,t+1}. \tag{115}$$

Here a_q, A^a, and G^a are the augmented counterparts of d_q, A_q, D_q, and G_q in Section 5, obtained by replacing y_t with $\widetilde{y}_t$ and appending the debt transition. Normalized transfers are selected from this state by

$$\widetilde{T}_t = c' q_{it}^a, \qquad c = (-\psi_y, 0, 1, 0)'. \tag{116}$$

Under the perceived law, the announced total consists of the observed current transfer, rule-implied future transfers, and planned future fiscal innovations:

$$s = c' q_{it}^a + \alpha_s + \bar{\theta}' q_{it}^a + \sum_{k=1}^{3} \omega_k \varepsilon_{t+k}^{\nu}. \tag{117}$$

Note that the above describes a planned allocation, and the countercyclical component of transfers is still expected by households to respond to fluctuations. The loading on the current state and the deterministic intercept are

$$\bar{\theta}' = \sum_{j=1}^{3} c' (A^a)^j, \qquad \alpha_s = \sum_{j=1}^{3} c' a^{(j)}, \tag{118}$$

where $a^{(j)}$ is the j-step transition intercept. The loading of the innovation in quarter $t+k$ on the cumulative package is

$$\omega_k = \left[(G^a)' \sum_{j=k}^{3} \left((A^a)^{j-k} \right)' c \right]_{\nu}. \tag{119}$$

The government knows the current state and its own fiscal plan, but not future productivity innovations. Future productivity shocks therefore do not enter (117). Any information about future productivity comes only from propagating the revised belief about the current state.

Numerically, the persistence loadings are

$$\omega = (2.703, 1.896, 0.999). \tag{120}$$

They are close to $(1 + \rho_\nu + \rho_\nu^2, 1 + \rho_\nu, 1)$. An innovation at $t+1$ affects all three future

quarters, an innovation at $t+2$ affects two quarters, and an innovation at $t+3$ affects only the final quarter. These are persistence loadings, not an exogenous quarterly allocation of the announced package.

Households already know the current transfer. Household i's announcement surprise is therefore

$$u_i = s - \widetilde{T}_t - \alpha_s - \bar{\theta}' m^a_{i,t|t} = s - E_{it}\left[\sum_{j=0}^{3} \widetilde{T}_{t+j}\right], \tag{121}$$

where $m^a_{i,t|t}$ denotes the posterior mean after observing the current wage, transfer, and tax payment but before receiving the cumulative announcement.

D.2 Joint Bayesian update

Collect the current state and the three future fiscal innovations in

$$x_i = \left((q^a_{it})', \varepsilon^{\nu}_{t+1}, \varepsilon^{\nu}_{t+2}, \varepsilon^{\nu}_{t+3}\right)'. \tag{122}$$

The prior mean and covariance are

$$E_{it}[x_i] = \begin{pmatrix} m^a_{i,t|t} \\ 0_3 \end{pmatrix}, \qquad \mathrm{Var}_{it}(x_i) = \begin{pmatrix} P & 0 \\ 0 & \sigma^2_{\nu} I_3 \end{pmatrix}. \tag{123}$$

The future fiscal innovations are independent before the announcement. Define

$$S = \bar{\theta}' P \bar{\theta} + \sigma^2_{\nu} \omega' \omega. \tag{124}$$

The posterior means are

$$E_{it}[q^a_{it} \mid s] = m^a_{i,t|t} + \frac{P\bar{\theta}}{S} u_i, \qquad E_{it}[\varepsilon^{\nu}_{t+k} \mid s] = \frac{\sigma^2_{\nu}\omega_k}{S} u_i. \tag{125}$$

After the current signals, uncertainty about the state is rank one:

$$P = p_y d d', \qquad d = (1, -1, \psi_y, 0)'. \tag{126}$$

The current transfer is constant along this remaining uncertainty direction because $c'd = 0$. Let

$$\lambda = \bar{\theta}' d = -0.0635. \tag{127}$$

The signal variance becomes

$$S = p_y \lambda^2 + \sigma^2_{\nu} \omega' \omega = 2.2 \times 10^{-6} + 1.54 \times 10^{-4} = 1.56 \times 10^{-4}. \tag{128}$$

The share of the announcement surprise attributed to uncertainty about the current state is

$$\varphi = \frac{p_y \lambda^2}{S} = 0.014, \qquad 1 - \varphi = 0.986. \tag{129}$$

Thus, households attribute 1.4% of the surprise to worse current conditions and 98.6% to the fiscal plan.

The output-belief update is

$$\Delta \widehat{\widetilde{y}}_{it} = \frac{p_y \lambda}{S} u_i = -0.219 u_i. \tag{130}$$

The posterior gains on the future fiscal innovations are

$$\big(E_{it}[\varepsilon^{\nu}_{t+1} \mid s], E_{it}[\varepsilon^{\nu}_{t+2} \mid s], E_{it}[\varepsilon^{\nu}_{t+3} \mid s]\big) = (0.224, 0.157, 0.083) u_i. \tag{131}$$

The inferred plan is front-loaded because an early innovation is more effective at generating a large cumulative package. Equivalently, the household selects the least statistically surprising sequence of future fiscal innovations consistent with the announced total. The contributions to the cumulative surprise are:

Table 17: Posterior decomposition of the announcement surprise

Explanation	Share of announcement surprise
Worse current economic state	1.4%
Planned innovation at $t+1$ and its persistent effects	60.5%
Planned innovation at $t+2$ and its persistent effects	29.8%
Planned innovation at $t+3$	8.3%

The contribution assigned to the $t+1$ innovation includes its persistent effects on transfers in $t+1$, $t+2$, and $t+3$. It should not be interpreted as the share paid in quarter $t+1$.

D.3 Mapping the update into GDP expectations

The model counterpart of the survey outcome is average expected output over the four-quarter window relative to Q1 2020:

$$G_i = \frac{100}{4} \sum_{j=0}^{3} E_{it}[\widetilde{y}_{t+j} \mid s]. \tag{132}$$

The treatment effect is linear in the household-specific surprise:

$$\Delta G_i = 100 \left(\Gamma_{\text{state}} + \Gamma_{\text{plan}}\right) u_i. \tag{133}$$

The current-state component is

$$\Gamma_{\text{state}} = \frac{1}{4}\sum_{j=0}^{3}\left[(A^a)^j \frac{P\bar{\theta}}{S}\right]_1 = -0.20704, \tag{134}$$

and the fiscal-plan component is

$$\Gamma_{\text{plan}} = \frac{1}{4}\sum_{j=1}^{3}\sum_{k=1}^{j}\left[(A^a)^{j-k} G_\nu^a\right]_1 \frac{\sigma_\nu^2 \omega_k}{S} = 0.00292. \tag{135}$$

Consequently,

$$\Delta G_i = -20.4118 u_i. \tag{136}$$

At the mean model surprise, $\bar{u} = 0.11199$, the model predicts

$$\overline{\Delta G} = -20.4118 \times 0.11199 = -2.286 \text{ percentage points}. \tag{137}$$

The empirical effect of -1.755 percentage points corresponds to a surprise of approximately $u = 0.086$. At this surprise,

$$\Delta\widehat{\widetilde{y}}_{it} = -0.219 \times 0.086 = -0.019, \tag{138}$$

and

$$\Delta G_{\text{state}} \approx -1.79 \text{ pp}, \qquad \Delta G_{\text{plan}} \approx 0.03 \text{ pp}, \qquad \Delta G \approx -1.76 \text{ pp}. \tag{139}$$

The bad-state inference lowers expected output in every quarter because the revision to current output is persistent. This does not represent news about additional future productivity shocks. The plan component raises expected output, but only slightly because fiscal innovations enter the perceived output law with a small coefficient. Under full information, the current state is already known, so the state component is absent and the announcement produces only the positive plan effect.

D.4 Conditional fiscal paths and consumption

The announcement fixes only one cumulative total. It does not identify the three future fiscal innovations separately. Let Σ_x denote the prior covariance of x_i and let g collect the corresponding signal loadings. The posterior covariance is

$$\text{Var}_{it}(x_i \mid s) = \Sigma_x - \frac{\Sigma_x g g' \Sigma_x}{g' \Sigma_x g}. \tag{140}$$

The individual quarterly innovations therefore remain uncertain and become negatively correlated after conditioning on the fixed total.

For a draw x_i^0 from the prior distribution, let $m_i^0 = E_{it}[x_i]$ denote its mean before the

announcement. The simulation constructs a treated draw by adding a projected, centered prior draw to the posterior mean:

$$x_i^T = E_{it}[x_i \mid s] + (x_i^0 - m_i^0) - \frac{\Sigma_x g}{g'\Sigma_x g} g'(x_i^0 - m_i^0). \tag{141}$$

Every treated draw can imply a different quarterly fiscal path, but every draw satisfies the same announced cumulative package. The household propagates the conditional path forward, observes future wages, transfers, and taxes, updates its beliefs, and applies their consumption policy.

D.5 Conditional output path

Figure 7 in Section 6.1 compares the model's conditional output distribution with realized German GDP. The productivity shock is chosen to reproduce the observed contraction between Q1 and Q2 2020. The realized recovery was substantially faster than the model's conditional path. The subsequent realized output path provides context for the exercise, while the validation targets households' expectations at the time of the announcement.

D.6 Mapping durable-purchase intentions into consumption revisions

This subsection describes the consumption-equivalent measure used in Section 6.1. Let ΔZ_h denote the treated-minus-untreated revision in expected Z_h. The horizon is $H = 4$ quarters, with $h = 1$ corresponding to Q2 2020 and $h = 4$ to Q1 2021. The inherited durable stock at the start of this window is fixed.

D.6.1 Purchase probabilities and net durable expenditure

For each category j, cars and other durables, let P_{0j} be the control-group purchase probability and G_{0j} baseline annual gross expenditure. Probabilities are expressed as fractions. Holding expenditure conditional on purchase fixed gives

$$Q_j = \frac{G_{0j}}{P_{0j}}, \qquad \Delta G_j = Q_j \Delta P_j. \tag{142}$$

Let η_j be the net-to-gross expenditure ratio, which deducts resale proceeds. Baseline net expenditure and its revision are

$$N_{0j} = \eta_j G_{0j}, \qquad \Delta N_j = \eta_j \Delta G_j. \tag{143}$$

The proportional revision in total net durable expenditure is therefore

$$R_B = \frac{\sum_j \Delta N_j}{\sum_j N_{0j}} = \sum_j w_j \frac{\Delta P_j}{P_{0j}}, \qquad w_j = \frac{N_{0j}}{\sum_k N_{0k}}. \tag{144}$$

The purchase probabilities and treatment effects are taken from the same BOP-HH estimation sample. Baseline expenditure inputs are from the German EVS 2018 reported by Destatis (2020).

D.6.2 Durable-stock accounting

Following the durable mapping in Laibson et al. (2022, Appendix B), let D_h denote the durable stock and x_h net durable purchases. The stock evolves according to

$$D_h = (1-\delta)D_{h-1} + x_h, \qquad D_h = \kappa_D c_h, \qquad \kappa_D = \frac{s_D}{r+\delta}, \tag{145}$$

where c_h is notional consumption, s_D the durable share, and $r+\delta$ the user cost. Prices, shares, and user costs are held fixed. With $\Delta D_0 = 0$, the revision in cumulative net durable purchases satisfies

$$\Delta X_H^D = \sum_{h=1}^{H} \Delta x_h = \Delta D_H + \delta \sum_{h=1}^{H-1} \Delta D_h = \kappa_D \left(\Delta c_H + \delta \sum_{h=1}^{H-1} \Delta c_h \right). \tag{146}$$

Thus, expenditure over the window reflects both the terminal stock adjustment and replacement purchases associated with the intervening stock revisions.

D.6.3 Consumption-equivalent revision

Let Δc_h^M be the model's quarterly consumption revision and $\bar{c}_0^M$ mean untreated consumption over the four quarters. Define

$$g_h^M = \frac{\Delta c_h^M}{\bar{c}_0^M}, \qquad \bar{g}^M = \frac{1}{H} \sum_{h=1}^{H} g_h^M. \tag{147}$$

Under stationary replacement and proportional stock adjustment by category,

$$N_{0j} = H\delta D_{0j}, \qquad \Delta D_{jh} = D_{0j} g_h^M. \tag{148}$$

The model-implied proportional revision in durable expenditure is

$$R_M = \frac{g_H^M + \delta \sum_{h=1}^{H-1} g_h^M}{H\delta}, \qquad \Delta P_j^M = P_{0j} R_M. \tag{149}$$

The consumption-equivalent path assumes a flat quarterly profile and rescales its magnitude to reproduce the empirical durable-expenditure revision:

$$g_h^{\mathrm{eq}} = \frac{R_B}{R_M} g_h^M, \qquad \bar{g}^{\mathrm{eq}} = \bar{g}^M \frac{R_B}{R_M}, \qquad R_M \neq 0. \tag{150}$$

The main table reports $100\bar{g}^{\mathrm{eq}}$ for the empirical consumption revision and $100\bar{g}^M$ for the model revision. The same conversion is applied to the heterogeneity comparisons.

Table 18: Calibration of the durable-to-consumption mapping

Parameter	Value
Horizon H	Four quarters
Quarterly depreciation δ	0.05
Annual gross expenditure on cars $G_{0,\mathrm{car}}$	EUR 1,452
Annual gross expenditure on other durables $G_{0,\mathrm{other}}$	EUR 876
Net-to-gross ratio for cars η_{car}	0.8
Net-to-gross ratio for other durables η_{other}	1
Net expenditure weights $(w_{\mathrm{car}}, w_{\mathrm{other}})$	$(0.570, 0.430)$

Notes: Expenditure inputs are from EVS 2018, Destatis (2020). The car adjustment deducts resale proceeds. Depreciation is a rounded benchmark from Laibson et al. (2022); housing is excluded from the mapping.

D.7 Household-income expectations

The survey elicits probabilities over brackets describing changes in monthly net household income over the following twelve months. The income concept includes after-tax labor income, pensions, and transfers. Table 19 reports the treatment effect on expected household net income, the model decomposition into labor income and transfers, and heterogeneity by current household income.

Table 19: Household-income expectations in the experiment and model

	Data	Dispersed information	Full information
Expected net-income revision	2.21	2.98	3.39
	(2.73)		
Of which: labor income	–	−0.29	0.35
Of which: transfers	–	3.28	3.04
Slope per EUR 1,000 of income	−1.23	−0.76	−0.75
	(2.00)		
Slope per income group	−0.74	−0.71	−0.70
	(2.07)		

Notes: The first row reports treatment effects on expected household net income, in percent. The decomposition gives contributions in percentage points. Slope rows report the change in the income revision, in percentage points, per EUR 1,000 or per one-category increase in current household income. Empirical standard errors are in parentheses. Components may differ from totals because of rounding.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations. Model calculations by author.

The positive but imprecise empirical income revision is consistent with both information structures. In the dispersed-information model, the contribution of expected labor income is negative, while expected transfers contribute positively and more than offset this decline. Under full information, both components are positive. The negative income gradients are similar across the two models and are imprecisely estimated in the data, so they do not distinguish between the information structures.

E Filtering with targeted transfers

This appendix describes the filtering method used to solve the targeted transfer model in Section 7. Targeting changes which continuous signals households observe and adds information through eligibility. The latter generates a non-Gaussian posterior even conditional on the linear perceived aggregate law. I retain a finite-dimensional belief state by projecting this posterior onto a Gaussian distribution with matching first and second moments. This follows assumed density filtering as described by Minka (2001).

E.1 State transition and continuous observations

For the targeted model, augment the baseline filtering state to include government debt:

$$q_{it} = (y_t, e_{it}, \nu_t, D_t)'. \tag{151}$$

Current debt is observed through taxes. Including it in the filtering state allows the prediction step to account for uncertainty about next-period debt among households that do not observe current aggregate transfers.

Under the perceived output law, the transition equations are

$$D_{t+1} = (R^* - \kappa)D_t + Y^{\text{ref}}\big[-\psi_y(y_t - y^{\text{ref}}) + \nu_t\big], \tag{152}$$

$$\nu_{t+1} = \rho_\nu \nu_t + \varepsilon^{\nu}_{t+1}, \tag{153}$$

$$e_{i,t+1} = \mu_\zeta + \rho_e e_{it} + \varepsilon^{e}_{i,t+1}, \tag{154}$$

$$y_{t+1} = b_0 + b_y y_t + b_B D_{t+1} + b_\nu \rho_\nu \nu_t + b_\mu \mu_{t+1} + b_\nu \varepsilon^{\nu}_{t+1}. \tag{155}$$

The innovations are mutually independent, with variances σ^2_ν, σ^2_ζ, and σ^2_μ, respectively. The output-law coefficients are estimated in the targeted economy.

Collecting these equations gives the transition equation

$$q_{i,t+1} = a^{\text{tar}} + F^{\text{tar}} q_{it} + G^{\text{tar}} \varepsilon_{i,t+1}, \tag{156}$$

where the matrices follow from equations (152)–(155). Let Q^{tar} denote the covariance matrix of the transition innovation. Given the previous period's Gaussian approximation, the prediction step is

$$m^-_{it} = a^{\text{tar}} + F^{\text{tar}} \hat{q}_{i,t-1|t-1}, \tag{157}$$

$$P^-_{it} = F^{\text{tar}} P_{i,t-1|t-1} (F^{\text{tar}})' + Q^{\text{tar}}. \tag{158}$$

Recipients observe their wage, the payment amount, and current debt. Since $T_{it} = 2T_t$ for recipients, their continuous observation vector is

$$s_{R,it} = \left(w_{it}, \frac{T_{it}}{2Y^{\text{ref}}}, D_t \right)'. \tag{159}$$

Nonrecipients observe their wage and current debt:

$$s_{N,it} = (w_{it}, D_t)'. \tag{160}$$

For $j \in \{R, N\}$, write the corresponding observation system as

$$s_{j,it} = h_j + H_j q_{it}. \tag{161}$$

The recipient matrices are

$$h_R = \begin{pmatrix} 0 \\ \psi_y y^{\text{ref}} \\ 0 \end{pmatrix}, \qquad H_R = \begin{pmatrix} 1 & 1 & 0 & 0 \\ -\psi_y & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{162}$$

while the nonrecipient matrices are

$$h_N = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad H_N = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{163}$$

First condition on the continuous observations, temporarily setting aside the inequality implied by eligibility. This gives the Gaussian intermediate distribution

$$\pi^G_{j,it} = \mathcal{N}(m^G_{j,it}, P^G_{j,it}), \tag{164}$$

with

$$K_{j,it} = P^-_{it} H_j' \left(H_j P^-_{it} H_j'\right)^\dagger, \tag{165}$$

$$m^G_{j,it} = m^-_{it} + K_{j,it}\left(s_{j,it} - h_j - H_j m^-_{it}\right), \tag{166}$$

$$P^G_{j,it} = P^-_{it} - K_{j,it} H_j P^-_{it}. \tag{167}$$

The generalized inverse allows for redundant observations that are already known exactly from the previous period's information.

This system also incorporates delayed learning from taxes. Once D_{t+1} is observed, the government budget constraint reveals

$$T_t = D_{t+1} - (R^* - \kappa) D_t. \tag{168}$$

Nonrecipients therefore recover the previous aggregate payment even though they do not observe it contemporaneously.

E.2 Eligibility and truncated posterior moments

Eligibility is determined by $e_{it} \leq \bar{e}$. Since $w_{it} = y_t + e_{it}$, this is equivalent to

$$y_t \geq k^y_{it}, \qquad k^y_{it} = w_{it} - \bar{e} = w_{it} + \frac{\sigma_e^2}{2}. \tag{169}$$

Thus, recipients truncate their intermediate distribution from below in output, while nonrecipients truncate it from above.

Let $a_y = (1, 0, 0, 0)'$ select aggregate output from the state vector. For each intermediate distribution, define

$$m^G_{y,j,it} = a_y' m^G_{j,it}, \tag{170}$$

$$s^2_{j,it} = a_y' P^G_{j,it} a_y, \tag{171}$$

$$\alpha_{j,it} = \frac{k^y_{it} - m^G_{y,j,it}}{s_{j,it}}. \tag{172}$$

Here and below, $s_{j,it}$ without bold notation denotes the scalar posterior standard deviation of output; the continuous observation vectors are those in equations (159) and (160).

Let ϕ and Φ denote the standard normal density and distribution functions. Conditional on the Gaussian prediction, the posterior incorporating both continuous observations and eligibility is

$$\pi_{it}(dq) = \begin{cases} \dfrac{\mathbf{1}\{a_y' q \geq k_{it}^y\}}{1 - \Phi(\alpha_{R,it})}\, \pi_{R,it}^G(dq), & \text{if } d_{it} = 1, \\ \dfrac{\mathbf{1}\{a_y' q < k_{it}^y\}}{\Phi(\alpha_{N,it})}\, \pi_{N,it}^G(dq), & \text{if } d_{it} = 0. \end{cases} \tag{173}$$

The denominators normalize the distributions after values inconsistent with observed eligibility have been removed.

The moments follow standard truncated-normal formulas (see e.g. Tallis, 1961). Define

$$\ell_{R,it} = \frac{\phi(\alpha_{R,it})}{1 - \Phi(\alpha_{R,it})}, \tag{174}$$

$$\ell_{N,it} = \frac{\phi(\alpha_{N,it})}{\Phi(\alpha_{N,it})}, \tag{175}$$

$$v_{j,it} = P_{j,it}^G a_y. \tag{176}$$

The recipient posterior has mean and covariance

$$\hat{q}_{it}^R = m_{R,it}^G + \frac{v_{R,it}}{s_{R,it}} \ell_{R,it}, \tag{177}$$

$$P_{it}^R = P_{R,it}^G - \frac{v_{R,it} v_{R,it}'}{s_{R,it}^2} \ell_{R,it} \left(\ell_{R,it} - \alpha_{R,it} \right). \tag{178}$$

For nonrecipients, the corresponding moments are

$$\hat{q}_{it}^N = m_{N,it}^G - \frac{v_{N,it}}{s_{N,it}} \ell_{N,it}, \tag{179}$$

$$P_{it}^N = P_{N,it}^G - \frac{v_{N,it} v_{N,it}'}{s_{N,it}^2} \ell_{N,it} \left(\ell_{N,it} + \alpha_{N,it} \right). \tag{180}$$

Eligibility shifts beliefs about all states correlated with output and reduces posterior uncertainty. Its informativeness depends on the position of the eligibility threshold relative to the intermediate output distribution. An eligibility outcome that was already very likely changes beliefs little. An outcome that removes a substantial part of the intermediate distribution produces a larger update. If the relevant output variance were zero, eligibility would already be determined and no truncation update needed.

E.3 Gaussian projection and household decisions

Following the assumed density filtering approach described by Minka (2001), I replace the truncated posterior by a Gaussian distribution with the same mean and covariance:

$$\pi_{it} \approx \mathcal{N}(\hat{q}_{it|t}, P_{it|t}), \tag{181}$$

where $(\hat{q}_{it|t}, P_{it|t})$ are selected from equations (177)–(180) according to eligibility. These moments are exact for the truncated distribution obtained from the incoming Gaussian approximation. The projection discards its higher moments and supplies the Gaussian input for the next prediction step.

Unlike in the baseline model, the covariance recursion depends on household observations through the standardized threshold. Posterior covariance matrices therefore differ across households and histories. Household decisions must account for these differences as well as differences in posterior means.

For example, under the projected posterior, the probability of eligibility next quarter is

$$\Pr_{it}(d_{i,t+1} = 1) = \Phi\left(\frac{\bar{e} - \mu_\zeta - \rho_e \hat{e}_{it|t}}{\sqrt{\rho_e^2 P_{ee,it|t} + \sigma_\zeta^2}}\right), \tag{182}$$

where $\hat{e}_{it|t}$ and $P_{ee,it|t}$ are the posterior mean and variance of current log efficiency. Both the mean and uncertainty about efficiency affect anticipated eligibility and thus the value of future support under the targeted rule.

For each candidate perceived output law, I solve household decisions using this filtering recursion. In the simulation, candidate aggregate consumption determines output, wages, aggregate transfers, and individual eligibility. Households observe the resulting signals, update their beliefs, and choose consumption. Aggregate consumption must equal the integral of these choices. The simulated output series then updates the perceived-law coefficients, as in the baseline solution method.

The targeted model therefore introduces two distinct approximations: the perceived law of aggregate output as in the baseline method, and now additionally the Gaussian representation of truncated household posteriors. The latter provides a tractable belief representation when eligibility makes exact conditional distributions non-Gaussian. Without this Gaussian projection or a similar approximation, repeated eligibility signals would generate non-Gaussian posteriors that cannot be summarized by their mean and covariance, requiring the full posterior distribution to enter the household state. This would make solving household consumption functions prohibitively costly under the global solution approach used in this paper.

E.4 Filtering accuracy

To evaluate the Gaussian approximation, I compare it with a reference Bayesian filter that retains the non-Gaussian distribution. Both filters receive the same simulated histories of wages, eligibility, payments, and debt. The comparison therefore evaluates the filtering approximation for a given simulated equilibrium.

Table 20 reports errors in output beliefs, posterior uncertainty, and predicted eligibility next quarter. Output-belief errors are normalized by the household's posterior standard deviation. The root mean squared error is 4.4% of that standard deviation across all observations, with a bias of 0.2%. Households' uncertainty about current output is approximately 2% of output, so the belief discrepancy is small relative to the uncertainty they face.

Table 20: Accuracy of the Gaussian filtering approximation

	Output belief		Posterior uncertainty	Eligibility
	RMSE	Bias	RMSE	RMSE
	(% of posterior s.d.)		(%)	(pp)
All observations	4.4	0.2	3.1	0.09
Near the cutoff (5%)	7.0	0.2	8.2	0.13

Notes: Gaussian approximation relative to reference Bayesian beliefs in the model targeting the bottom half of the income distribution. RMSE denotes root mean squared error across households and quarters. Output-belief errors and bias are expressed as percentages of the household's posterior standard deviation of output. Errors in posterior uncertainty are reported in percent, and errors in next-quarter eligibility probabilities in percentage points. The near-cutoff group consists of observations for which the eligibility threshold lies within one posterior standard deviation of the household's output belief, approximately 5% of observations.

Errors are larger near the eligibility cutoff, where the binary signal is particularly informative. Even in this group, the root mean squared output-belief error is 7.0% of posterior uncertainty. The error in the probability of eligibility next quarter is 0.09 percentage points overall and 0.13 percentage points near the cutoff.

The reported discrepancies arise from the recursive Gaussian approximation. Matching moments reproduces the mean and covariance of each truncated Gaussian exactly, but subsequent predictions start from the projected distribution rather than the full non-Gaussian history. These checks assess that additional approximation separately from the approximation of aggregate dynamics by the perceived law of motion.